

Prediction-Powered Data Fusion for Treatment Effect Estimation

Yonghan Jung

Working draft, September 9, 2026

Randomized controlled trials identify causal effects under transparent design assumptions, but their sample sizes can make average and conditional treatment-effect estimation imprecise. Observational studies are often larger and can yield accurate outcome predictions, but those predictions need not be causal. We develop a model-agnostic prediction-powered framework that keeps the randomized trial as the causal anchor while using observational outcome models and covariates through two tunable channels. The coefficient λ changes the outcome regression inside an RCT-valid pseudo-outcome, and ω introduces a prediction-powered control variate centered across the two samples.

For the average treatment effect, we derive exact conditional unbiasedness, a within-/between-covariate variance decomposition, the joint population variance minimizer, its separable specialization, singular and constrained cases, and an honest plug-in algorithm. A conditional central limit theorem supports normal-approximation inference when nuisance fitting, coefficient tuning, and final evaluation use independent data blocks; the present theory does not cover cross-fitted sample reuse. For conditional effects, we formulate RCT-valid DR-Fusion and R-Fusion losses, audit their nuisance sensitivity, and develop general two-source learning bounds with linear-sieve and additive-spline specializations. Exact density-ratio transport covers unequal covariate marginals; classifier odds, Riesz learning, and loss-targeted balancing enter an estimated-weight regime with explicit drift. Existing simulations support the ATE construction in several small-trial regimes. CATE experiments and real-outcome analyses remain planned, so neither the theory nor the current evidence establishes universal finite-sample improvement for estimated coefficients.

Contents

1. Introduction	3
1.1. Categorized literature review	3
2. Background and Problem Setting	4
2.1. RCT-OBS problem setting and notation	4
2.2. Existing data-fusion estimators	7

3. Prediction-Powered Fusion for the ATE	8
3.1. Estimator, exact variance, and population oracles	8
3.2. Estimable honest algorithm	10
3.2.1. Coefficient estimation and honest evaluation	11
3.2.2. Estimating the density ratio	13
4. Data Fusion for CATE Estimation	18
4.1. CATE targets and RCT-valid learning losses	18
4.2. Orthogonality of learning equations and nuisance sensitivity	20
4.3. Fitting the learners and choosing the fusion coefficients	21
4.4. Linear-sieve and additive-spline specialization	23
4.5. Implementation boundaries and open theory	28
5. Experiments	28
5.1. Completed ATE benchmarks	28
5.2. Planned CATE evaluation and interpretation	30
5.3. Scope, failure modes, and conclusion	30
A. Omitted Preliminaries	34
A.1. Prediction-powered inference and PPI^{++}	34
B. Omitted Results	35
B.1. Convexity of the fusion losses	35
B.2. Closed-form noise-floor coefficients	35
C. Simulation Details	36
D. Proof Details	38
D.1. Conditional validity of the trial pseudo-outcome	38
D.2. Proof of Corollary 1	38
D.3. Proof of Theorem 1	38
D.4. Proof of Corollary 1.1	39
D.5. Proof of Proposition 2	39
D.6. Proof of Theorem 2	39
D.7. Compact constraints	40
D.8. Proof of Corollary 2.1	40
D.9. Prediction-powered CATE losses and finite-sample theory	40
D.10. Proofs for coefficient selection	40
D.11. Proofs for the orthogonality audit	41
D.12. A two-stratum oracle calculation	43
D.13. Honest variance estimation and evaluation-sample inference	43
D.14. Proof of Theorem 3	44
D.15. Density-ratio characterizations and estimated-ratio inheritance	44
D.16. Counterexamples to adaptive reuse	47

1. Introduction

1.1. Categorized literature review

Prediction-powered inference. Prediction-powered inference uses a large set of machine predictions together with a smaller labeled sample, correcting the prediction-based estimate with labeled residuals so that the statistical target does not require the predictor to be correct (Angelopoulos et al., 2023). This separation between predictive quality and inferential validity is the starting point for the present construction. Randomized observations play the role of causally valid labeled data, while observational outcome models supply potentially useful predictions.

Adaptive prediction-powered inference. PPI⁺⁺ introduces data-adaptive power tuning that adjusts how strongly predictions enter an estimating equation and reports computational and statistical gains over the original construction (Angelopoulos et al., 2024). Its central lesson is that a prediction should be used in proportion to its empirical value rather than being accepted as truth. Our coefficient ω follows this control-variate logic, but the target and score are causal quantities identified by the randomized design.

Causal data fusion and transport. General causal data-fusion theory distinguishes data collected under different regimes and populations and asks whether a target causal query is identified from their combination (Bareinboim and Pearl, 2016). Reviews of causal data integration likewise separate randomized trials, observational studies, historical controls, and heterogeneous target populations (Shi et al., 2023). The present paper assumes that the trial already identifies the target effect and studies an estimation question: how can observational predictions improve precision without replacing randomized identifying information?

Combining randomized and observational estimates. Existing estimators combine experimental and observational effect estimates through shrinkage, bias detection, or adaptive weighting. Shrinkage estimators can reduce risk when observational bias is sufficiently controlled (Rosenman et al., 2023), while adaptive combination methods move toward the randomized estimator when observational bias is detected (Cheng and Cai, 2021). These approaches combine effect estimates from the two sources. Our construction instead inserts the observational outcome model into an RCT-valid pseudo-outcome and separately centers a prediction-based covariate statistic.

CATE learners and orthogonal losses. Meta-learners convert standard regression algorithms into conditional effect estimators (Künzel et al., 2019). The R-learner residualizes outcomes and treatments to isolate treatment-effect heterogeneity (Nie and Wager, 2021), while doubly robust pseudo-outcome regression can retain favorable error structure under nuisance estimation (Kennedy, 2023). Direct multi-source learners have also been proposed for heterogeneous causal data fusion (Li et al., 2022). These results motivate treating the prediction-powered construction as a loss wrapper that can be paired with a linear sieve or a fixed learned representation.

Covariate shift and balancing. When trial and observational covariate marginals differ, exact density-ratio weighting transports fixed covariate functions between populations. Riesz-representer methods provide an optimization perspective on such linear functionals and on debiasing (Chernozhukov et al., 2021). Density-ratio estimation, generic balancing, and Riesz learning are established tools. The project-specific question is whether they can be targeted to the generated CATE loss class $\{(\hat{g} - t)^2 : t \in \mathcal{T}\}$, where $\hat{g}$ is the fitted OBS treatment-effect prediction defined later in Equation (14), while preserving an RCT-defined risk; we treat this specialization as a working

research direction rather than a settled novelty claim.

2. Background and Problem Setting

2.1. RCT–OBS problem setting and notation

Let $\mathcal{K} \triangleq \{\text{eval}, \text{nuis}, \text{tune}\}$ index the three statistical roles assigned to the samples. For each source $S \in \{R, O\}$, where R and O denote the randomized and observational sources, respectively, and each role $k \in \mathcal{K}$, define

$$D_S^k \triangleq \{V_{S,i}^k = (X_{S,i}^k, A_{S,i}^k, Y_{S,i}^k) : i = 1, \dots, n_S^k\} \sim P_S^{\otimes n_S^k}, \quad (1)$$

as the sample from source S reserved for role k . Here X takes values in the covariate space $\mathcal{X}$, $A \in \{0, 1\}$ is treatment, and Y is the observed outcome. For each $S \in \{R, O\}$, the datasets D_S^{eval} , D_S^{nuis} , and D_S^{tune} are used for final evaluation, nuisance and density-ratio fitting, and coefficient or candidate tuning, respectively.

We impose the following independence condition across the six role-specific samples.

Assumption 1 (Honest sample separation). The six datasets $\{D_S^k : S \in \{R, O\}, k \in \mathcal{K}\}$ are mutually independent.

Set $n \triangleq n_R^{\text{eval}}$ and $N \triangleq n_O^{\text{eval}}$, and define $\mathbb{P}_{R,n}h \triangleq n^{-1} \sum_i h(V_{R,i}^{\text{eval}})$ and $\mathbb{P}_{O,N}h \triangleq N^{-1} \sum_j h(V_{O,j}^{\text{eval}})$. Every displayed moment is finite; square-integrability and stronger moment conditions are stated locally when variance estimation or normal-approximation inference requires them.

Let $Y(1)$ and $Y(0)$ denote potential outcomes. The RCT-population conditional and average treatment effects are

$$\tau(x) \triangleq \mathbb{E}_R\{Y(1) - Y(0) \mid X = x\}, \quad (2)$$

$$\theta \triangleq \mathbb{E}_R\{\tau(X)\}. \quad (3)$$

For causal identification in the trial population, we impose the following assumptions under P_R .

Assumption 2 (Trial identification). The following assumptions hold:

1. $Y = Y(A)$ almost surely;
2. $\{Y(0), Y(1)\} \perp A \mid X$ under the interventional distribution P_R ;
3. The trial propensity $e(x) \triangleq P_R(A = 1 \mid X = x)$ is known and satisfies the positivity: $\epsilon < e(X) < 1 - \epsilon$ almost surely for a constant $\epsilon > 0$.

We study both the common-covariate regime $P_R^X = P_O^X$, formalized in Assumption 3, and the more general covariate-shift regime $P_R^X \neq P_O^X$, formalized in Assumption 4.

Definition 1 (Target-to-source density ratio). Whenever $P_R^X \ll P_O^X$, define the target-to-source density ratio by

$$r_0(x) \triangleq \frac{dP_R^X}{dP_O^X}(x).$$

It is the Radon–Nikodym derivative that transports an OBS covariate expectation to the RCT covariate population.

Assumption 3 (Common covariate marginal). The trial and observational covariates have the same marginal law, $P_R^X = P_O^X$. This assumption does not equate the treatment or outcome mechanisms across sources, and the observational treatment contrast may remain confounded. In this regime, Definition 1 gives $r_0(x) = 1$ for P_O^X -almost every x .

Assumption 4 (Covariate shift and support overlap). Under covariate shift, $P_R^X \ll P_O^X$, so the RCT covariate population has no support outside the OBS covariate population. Definition 1 therefore defines r_0 . This assumption establishes existence of the ratio; it does not assert that r_0 is known or exactly estimated.

Assumptions 3 and 4 describe alternative covariate regimes, not cumulative restrictions. Under Assumption 3, the exact density ratio is $r_0 = 1$, so no transport weighting is needed. When the covariate marginals differ, Assumption 4 is needed to convert OBS covariate expectations into expectations under the RCT population.

For $a \in \{0, 1\}$, define the RCT outcome-regression target $\mu_{R,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{R,a}(x) \triangleq \mathbb{E}_R[Y \mid X = x, A = a].$$

For each $a \in \{0, 1\}$, let $\hat{\mu}_{R,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of $\mu_{R,a}$ trained only on D_R^{nuis} . Define

$$\hat{\mu}_R \triangleq (\hat{\mu}_{R,0}, \hat{\mu}_{R,1}). \quad (4)$$

We define the following AIPW score.

Definition 2 (RCT-valid AIPW score). Let $q \triangleq (q_0, q_1)$ be a fixed pair of measurable functions from $\mathcal{X}$ to $\mathbb{R}$. Define

$$\varphi(V; q) \triangleq q_1(X) - q_0(X) + \frac{A}{e(X)}\{Y - q_1(X)\} - \frac{1 - A}{1 - e(X)}\{Y - q_0(X)\}. \quad (5)$$

This score gives a valid causal estimate:

Proposition 1 (RCT-validity for an arbitrary outcome regression). Suppose Assumption 2 holds. Let q be a measurable function pair fixed independently of the RCT evaluation sample, and suppose $\varphi(V; q)$ is integrable. Then

$$\mathbb{E}_R\{\varphi(V; q) \mid X\} = \tau(X). \quad (6)$$

Definition 3 (RCT-only AIPW estimator). Define the RCT-only AIPW estimator on the evaluation sample by

$$\hat{\theta}_{\text{AIPW},R} \triangleq \mathbb{P}_{R,n}\varphi(V; \hat{\mu}_R). \quad (7)$$

Corollary 1 (RCT-only AIPW recovery). Suppose Assumptions 2 and 1 hold and $\varphi(V; \hat{\mu}_R)$ is integrable. Conditional on the nuisance data used to fit $\hat{\mu}_R$, $\mathbb{E}\{\hat{\theta}_{\text{AIPW},R} \mid D_R^{\text{nuis}}\} = \theta$.

For $a \in \{0, 1\}$, define the OBS outcome-regression target $\mu_{O,a} : \mathcal{X} \rightarrow \mathbb{R}$ by

$$\mu_{O,a}(x) \triangleq \mathbb{E}_O[Y \mid X = x, A = a].$$

For each $a \in \{0, 1\}$, let $\hat{\mu}_{O,a} : \mathcal{X} \rightarrow \mathbb{R}$ denote a fixed pre-evaluation predictor of $\mu_{O,a}$ trained only on D_O^{nuis} . Define

$$\hat{\mu}_O \triangleq (\hat{\mu}_{O,0}, \hat{\mu}_{O,1}). \quad (8)$$

The RCT-validity result also permits the RCT and OBS outcome regressions to be combined without changing the conditional causal target.

Definition 4 (Fusion score). For $\lambda \in \mathbb{R}$, define

$$\hat{\mu}_\lambda \triangleq (1 - \lambda)\hat{\mu}_R + \lambda\hat{\mu}_O, \quad (9)$$

and the fused score as

$$Z_\lambda \triangleq \varphi(V; \hat{\mu}_\lambda). \quad (10)$$

From $Z_0 = \varphi(V; \hat{\mu}_R)$, $Z_1 = \varphi(V; \hat{\mu}_O)$ and $\Delta \triangleq Z_1 - Z_0$, we can rewrite the fused score as

$$Z_\lambda = Z_0 + \lambda\Delta. \quad (11)$$

Section 4 asks how a fitted nuisance enters the first-order condition of a loss. The following two definitions fix the language.

Definition 5 (Gateaux derivative (van der Vaart, 1998, Chapter 20)). Let F be a functional of a function-valued argument η , and let h be a direction such that $\eta + \varepsilon h$ stays in the domain of F for all sufficiently small $|\varepsilon|$. Whenever the limit exists, the Gateaux derivative of F at

η in the direction h is

$$D_\eta F(\eta)[h] \triangleq \left. \frac{d}{d\varepsilon} F(\eta + \varepsilon h) \right|_{\varepsilon=0} \quad (12)$$

Definition 6 (Learning equation and orthogonality (Chernozhukov et al., 2018; Foster and Syrgkanis, 2023)). Let $L(\zeta; \eta)$ be a population loss with a target argument ζ in a linear class and a nuisance argument η , and let η_0 denote the true nuisance. The learning equation of L is

$$\Psi(\zeta; \eta)[v] \triangleq -\frac{1}{2} D_\zeta L(\zeta; \eta)[v] \quad (13)$$

for target directions v in the class. The loss is orthogonal to η if $D_\eta \Psi(\zeta^*; \eta_0)[h, v] = 0$ for every admissible nuisance direction h and every target direction v .

When L is convex in ζ , a function ζ^* minimizes $L(\cdot; \eta_0)$ over the class exactly when $\Psi(\zeta^*; \eta_0)[v] = 0$ for every v in the class, which is why Ψ is called the learning equation.

2.2. Existing data-fusion estimators

Data integration first asks whether information from different regimes or populations identifies a common target. Transport and general data-fusion frameworks make the source and target populations explicit (Bareinboim and Pearl, 2016; Shi et al., 2023; Colnet et al., 2020). Our target is already identified by the RCT; transport is used to align fixed OBS covariate functions with the RCT population.

Robust borrowing methods instead trade potential efficiency gains against observational bias. They include shrinkage, compatibility testing, and adaptive weighting that returns toward the randomized estimate when the sources disagree (Rosenman et al., 2023; Cheng and Cai, 2021). These methods clarify why observational information can help without implying a universal gain.

For heterogeneous effects, direct multi-source learners combine weighted pseudo-outcomes, while experimental-grounding and RCT–OBS procedures use the trial to constrain or diagnose observational HTE estimates (Li et al., 2022; Kallus et al., 2018). The elastic procedure of Yang et al. (2023) tests compatibility before pooling real-world data for prespecified effect modifiers. The confounding-function approach of Yang et al. (2025) jointly models an HTE function and an observational confounding function under explicit identification conditions.

Our construction differs at the fusion object. The coefficient λ changes the outcome regression inside an RCT-valid score, whereas the second coefficient scales a transported mean-zero control variate built from a fitted OBS treatment-effect prediction. This PPI-style use of prediction does not combine two causal effect estimates and does not require the OBS contrast to equal the CATE.

3. Prediction-Powered Fusion for the ATE

Throughout this section, Assumptions 2 and 1 hold, the evaluation sample sizes satisfy $n, N > 0$, and either Assumption 3 or Assumption 4 holds. Thus, the exact density ratio r_0 in Definition 1 exists and equals one under the common-covariate regime. A feasible estimator may use a fixed candidate weight r ; results that require the exact weight state $r = r_0$ explicitly.

3.1. Estimator, exact variance, and population oracles

Definition 7 (ATE fusion estimator and its components). Using the fitted OBS outcome regression $\hat{\mu}_O$ from Equation (8), define the fitted observational treatment-effect prediction by

$$\hat{g}(x) \triangleq \hat{\mu}_{O,1}(x) - \hat{\mu}_{O,0}(x). \quad (14)$$

Let $r : \mathcal{X} \rightarrow [0, \infty)$ be a fixed measurable candidate transport weight. For fixed (λ, ω) , define the ATE fusion estimator by

$$\hat{\theta}_r(\lambda, \omega) \triangleq \mathbb{P}_{R,n} Z_\lambda + \omega \{ \mathbb{P}_{O,N} \{ r(X) \hat{g}(X) \} - \mathbb{P}_{R,n} \hat{g}(X) \}, \quad (15)$$

where Z_λ is the RCT-valid score in Equation (10). The fitted outcome regressions $\hat{\mu}_R$ and $\hat{\mu}_O$ are defined in Equations (4) and (8), respectively. The coefficient λ blends these regressions inside Z_λ , whereas ω scales the transported control variate. Exact transport uses $r = r_0$.

At the coefficient origin,

$$\hat{\theta}_r(0, 0) = \mathbb{P}_{R,n} Z_0 = \hat{\theta}_{\text{AIPW},R},$$

where $\hat{\theta}_{\text{AIPW},R}$ is defined in Equation (7), so the family contains the RCT-only AIPW estimator.

The estimator depends on three learned components: the outcome-regression blend inside $Z_\lambda \triangleq \varphi(V; \hat{\mu}_\lambda)$, the OBS prediction $\hat{g}$, and the density ratio r . The next result states exactly which components can change its population mean.

Theorem 1 (Exact target, density-ratio drift, and variance). For every fixed (λ, ω) , arbitrary fixed outcome-regression fits and a fixed OBS prediction satisfy, whenever the displayed expectations are finite,

$$\mathbb{E}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \theta. \quad (16)$$

For any fixed candidate weight $\tilde{r}$, its exact mean drift is

$$\mathbb{E}\{\hat{\theta}_{\tilde{r}}(\lambda, \omega)\} - \theta = \omega \mathbb{E}_O [\{\tilde{r}(X) - r_0(X)\} \hat{g}(X)]. \quad (17)$$

If $Z_\lambda - \omega \hat{g}(X)$ and $r_0(X)\hat{g}(X)$ are square-integrable, then

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R\{Z_\lambda - \omega \hat{g}(X)\} + \frac{\omega^2}{N} \text{Var}_O\{r_0(X)\hat{g}(X)\}. \quad (18)$$

Equation (16) shows that, under exact transport $r = r_0$, the expectation of $\hat{\theta}_{r_0}(\lambda, \omega)$ is unchanged by the fixed outcome regressions $\hat{\mu}_R$ and $\hat{\mu}_O$ or by the fixed OBS prediction $\hat{g}$. If a candidate weight $\tilde{r}$ differs from r_0 , Equation (17) gives the resulting mean error exactly.

We next minimize the exact variance in Equation (18) over the two fusion coefficients.

Definition 8 (ATE variance coefficients). Recall from Equation (10) and Section 2.1 that $Z_0 \triangleq \varphi(V; \hat{\mu}_R)$ and $\Delta \triangleq Z_1 - Z_0$; $\hat{g}$ is the OBS prediction in Equation (14). Define

$$\begin{aligned} A &\triangleq \text{Var}_R(\Delta), & B &\triangleq \text{Var}_R(\hat{g}) + \frac{n}{N} \text{Var}_O(r_0 \hat{g}), \\ C &\triangleq \text{Cov}_R(Z_0, \Delta), & D &\triangleq \text{Cov}_R(Z_0, \hat{g}). \end{aligned} \quad (19)$$

Here A and B are the quadratic costs of changing λ and ω , while C and D determine their linear slopes. Also define

$$\mathbf{H}_{\text{ATE}} \triangleq \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}, \quad \boldsymbol{\ell} \triangleq \begin{pmatrix} C \\ -D \end{pmatrix}, \quad \boldsymbol{\beta} \triangleq \begin{pmatrix} \lambda \\ \omega \end{pmatrix}.$$

The following corollary rewrites the variance in Equation (18) using the four coefficients A , B , C , and D .

Corollary 1.1 (Separated ATE variance quadratic). If the displayed scores have finite second moments, then

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \left\{ \text{Var}_R(Z_0) + 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2 \right\}. \quad (20)$$

Removing the constant term from Equation (20) gives the following equivalent optimization objective.

Proposition 2 (Equivalent ATE coefficient objective). Define

$$\mathcal{J}_{\text{ATE}}(\lambda, \omega) \triangleq 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2.$$

For every nonempty compact $\Theta \subseteq \mathbb{R}^2$,

$$\arg \min_{(\lambda, \omega) \in \Theta} \text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \arg \min_{(\lambda, \omega) \in \Theta} \mathcal{J}_{\text{ATE}}(\lambda, \omega).$$

The next theorem gives the closed-form variance-minimizing coefficients and their ridge-regularized counterparts.

Theorem 2 (Population and ridge oracles). Suppose the displayed scores have finite second moments and $A, B > 0$. For every $\rho \geq 0$, the unique ridge optimizer is

$$\begin{pmatrix} \lambda_\rho^* \\ \omega_\rho^* \end{pmatrix} \triangleq \arg \min_{(\lambda, \omega) \in \mathbb{R}^2} \left\{ \mathcal{J}_{\text{ATE}}(\lambda, \omega) + \rho(\lambda^2 + \omega^2) \right\} = \begin{pmatrix} -C/(A + \rho) \\ D/(B + \rho) \end{pmatrix}. \quad (21)$$

At $\rho = 0$, this is the unique minimizer of the exact population variance:

$$\lambda^* \triangleq \lambda_0^* = -\frac{C}{A}, \quad \omega^* \triangleq \omega_0^* = \frac{D}{B}. \quad (22)$$

Corollary 2.1 (Minimum population variance). Suppose the displayed scores have finite second moments and $A, B > 0$. Define

$$\mathcal{V}^* \triangleq \min_{(\lambda, \omega) \in \mathbb{R}^2} \text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\}. \quad (23)$$

Let $\mathcal{V}_0 \triangleq \text{Var}_R(Z_0)/n$. Then

$$\mathcal{V}^* = \mathcal{V}_0 - \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right), \quad \mathcal{V}_0 - \mathcal{V}^* = \frac{1}{n} \left(\frac{C^2}{A} + \frac{D^2}{B} \right) \geq 0. \quad (24)$$

The next theorem separates the two sources of this oracle variance reduction.

Theorem 3 (Exact-ratio oracle variance comparison). Suppose the scores in Equation (18) have finite second moments and $A, B > 0$. Define $\tilde{\mathcal{V}}(\lambda, \omega) \triangleq n \text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\}$ and $\sigma_0^2 \triangleq \text{Var}_R(Z_0)$. Here “PPI⁺⁺-only” means the member of our family that fixes $\lambda = 0$ and optimally selects ω . Then

$$\begin{aligned} \tilde{\mathcal{V}}_{\text{RCT}} &\triangleq \tilde{\mathcal{V}}(0, 0) = \sigma_0^2, \\ \tilde{\mathcal{V}}_{\text{PPI}^{++}} &\triangleq \tilde{\mathcal{V}}(0, D/B) = \sigma_0^2 - \frac{D^2}{B}, \\ \tilde{\mathcal{V}}_{\text{joint}} &\triangleq \tilde{\mathcal{V}}(-C/A, D/B) = \sigma_0^2 - \frac{D^2}{B} - \frac{C^2}{A}. \end{aligned} \quad (25)$$

Therefore $\tilde{\mathcal{V}}_{\text{joint}} \leq \tilde{\mathcal{V}}_{\text{PPI}^{++}} \leq \tilde{\mathcal{V}}_{\text{RCT}}$.

Thus, with population-oracle coefficients and the exact density ratio, the joint ATE estimator has variance no larger than either the PPI⁺⁺-only estimator or the RCT-only estimator.

3.2. Estimable honest algorithm

This subsection gives a finite-sample algorithm for estimating the two fusion coefficients.

We first define the feasible region over which λ and ω are selected.

Definition 9 (Coefficient rectangle and interval projection). Define the nonempty compact coefficient rectangle by

$$\Theta_{\text{rect}} \triangleq [L_\lambda, U_\lambda] \times [L_\omega, U_\omega]. \quad (26)$$

For a closed interval $[L, U]$ with $L \leq U$, define

$$\text{Proj}_{[L, U]}(u) \triangleq \min\{U, \max(L, u)\}. \quad (27)$$

This operator returns u when it lies in the interval and otherwise returns the nearest endpoint.

For positive A, B , the separated objective gives the coordinatewise projection

$$\lambda_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\lambda, U_\lambda]}(-C/A), \quad \omega_{\Theta_{\text{rect}}}^* = \text{Proj}_{[L_\omega, U_\omega]}(D/B).$$

The population oracle weakly improves the variance objective whenever $(0, 0)$ is feasible. Estimated coefficients have no universal finite-sample no-harm guarantee.

3.2.1. Coefficient estimation and honest evaluation

For $S \in \{R, O\}$, suppose $n_S^{\text{tune}} \geq 2$. Conditional on the fixed nuisance fits, $\widehat{\text{Cov}}_S$ and $\widehat{\text{Var}}_S$ denote the usual unbiased tuning-sample covariance and variance with denominator $n_S^{\text{tune}} - 1$. The factor n/N below uses the evaluation-sample sizes whose variance is optimized.

Let $\hat{r}$ denote a density-ratio estimator fitted only on the nuisance samples. Under the common-covariate regime, set $\hat{r} \triangleq 1$.

Using the tuning samples, compute the following four empirical analogues of A , B , C , and D in Definition 8:

$$\begin{aligned} \hat{A} &\triangleq \widehat{\text{Var}}_R(\Delta), & \hat{B} &\triangleq \widehat{\text{Var}}_R(\hat{g}) + \frac{n}{N} \widehat{\text{Var}}_O(\hat{r}\hat{g}), \\ \hat{C} &\triangleq \widehat{\text{Cov}}_R(Z_0, \Delta), & \hat{D} &\triangleq \widehat{\text{Cov}}_R(Z_0, \hat{g}). \end{aligned} \quad (28)$$

These four moments define the separated empirical objective

$$\hat{\mathcal{J}}_{\text{ATE}}(\lambda, \omega) \triangleq 2\hat{C}\lambda - 2\hat{D}\omega + \hat{A}\lambda^2 + \hat{B}\omega^2. \quad (29)$$

Before observing the tuning outcomes, prespecify the bounds defining the nonempty compact rectangle Θ_{rect} , require it to contain the origin, and choose positive thresholds t_A and t_B .

Recall that $\hat{\mu}_R = (\hat{\mu}_{R,0}, \hat{\mu}_{R,1})$ and $\hat{\mu}_O = (\hat{\mu}_{O,0}, \hat{\mu}_{O,1})$ are outcome-regression predictors fitted on the RCT and OBS nuisance samples, respectively, and $\hat{g}$ is their fitted OBS contrast from Equation (14). Algorithm 1 states the independent-sample implementation covered by the exact finite-sample results. The nuisance, tuning, and evaluation samples may be supplied directly or obtained by an honest prespecified split.

Algorithm 1: ATE estimation with honest evaluation

Input: Nuisance samples $D_R^{\text{nuis}}, D_O^{\text{nuis}}$, tuning samples $D_R^{\text{tune}}, D_O^{\text{tune}}$, evaluation samples $D_R^{\text{eval}}, D_O^{\text{eval}}$, the rectangle Θ_{rect} , and denominator thresholds.

- 1 Fit $\hat{\mu}_R, \hat{\mu}_O, \hat{g}$, and a density-ratio estimator $\hat{r}$ on $D_R^{\text{nuis}}, D_O^{\text{nuis}}$. The function $\hat{r}$ estimates the exact density ratio r_0 . When $\hat{r} \neq r_0$, Equation (17) gives the resulting mean error.
- 2 Compute the four moments in Equation (28) on $D_R^{\text{tune}}, D_O^{\text{tune}}$, and select the coefficients using the threshold-and-projection rule in Equation (30).
- 3 Set $r = \hat{r}$ in Equation (15), and evaluate it on $D_R^{\text{eval}}, D_O^{\text{eval}}$. Set $\psi_R \triangleq Z_{\hat{\lambda}} - \hat{\omega}\hat{g}$ and $\psi_O \triangleq \hat{\omega}\hat{r}\hat{g}$. Compute $\hat{V} \triangleq \widehat{\text{Var}}_{R,\text{eval}}(\psi_R)/n + \widehat{\text{Var}}_{O,\text{eval}}(\psi_O)/N$.

Output: $\hat{\theta}_r$ and $\hat{V}$.

How Algorithm 1 selects the coefficients. Because Equation (29) is separated in λ and ω , Algorithm 1 selects the two coefficients coordinatewise and projects each one onto its prespecified interval:

$$\hat{\lambda} \triangleq \begin{cases} \text{Proj}_{[L_{\lambda}, U_{\lambda}]}(-\hat{C}/\hat{A}), & \hat{A} > t_A, \\ 0, & \hat{A} \leq t_A, \end{cases} \quad \hat{\omega} \triangleq \begin{cases} \text{Proj}_{[L_{\omega}, U_{\omega}]}(\hat{D}/\hat{B}), & \hat{B} > t_B, \\ 0, & \hat{B} \leq t_B. \end{cases} \quad (30)$$

Analysis. The four-moment rule in Equation (30) directly estimates the two population-oracle coordinates in Equation (22). The thresholds avoid division by nearly zero empirical curvature, and the projections enforce the prespecified feasible rectangle.

For Algorithm 1, set $r = \hat{r}$ in the generic evaluation scores below. For explicit evaluation notation, define

$$\psi_{R,i} \triangleq Z_{\hat{\lambda}}(V_{R,i}^{\text{eval}}) - \hat{\omega}\hat{g}(X_{R,i}^{\text{eval}}), \quad \psi_{O,j} \triangleq \hat{\omega}r(X_{O,j}^{\text{eval}})\hat{g}(X_{O,j}^{\text{eval}}),$$

Let $\widehat{\text{Var}}_{R,\text{eval}}$ and $\widehat{\text{Var}}_{O,\text{eval}}$ be the usual unbiased sample variances with denominators $n - 1$ and $N - 1$, respectively. Define the fixed-selected-coefficient variance estimator by

$$\hat{V} \triangleq \frac{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)}{n} + \frac{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)}{N}. \quad (31)$$

Proposition 3 (Exact variance estimation after honest selection). Let $\hat{\theta}_r$ and $\hat{V}$ be the estimator and variance estimator from Algorithm 1, specialized to $r = \hat{r} = r_0$. Suppose Assumption 1 holds, $n, N \geq 2$, and $\psi_R \in L_2(P_R)$, $\psi_O \in L_2(P_O)$. Conditional on all fitted and selected objects,

$$\mathbb{E}(\hat{\theta}_r) = \theta, \quad \mathbb{E}(\hat{V}) = \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\}.$$

When the corresponding unconditional moments exist, the same identities also hold unconditionally because the conditional mean is the nonrandom target θ .

Theorem 4 (Normal approximation after honest evaluation). Let $\hat{\theta}_r$ and $\hat{V}$ be the estimator and variance estimator from Algorithm 1, specialized to $r = \hat{r} = r_0$. Suppose Assumption 1 holds, $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and their centered scores have uniformly bounded conditional $(2 + \delta)$ -moments for some $\delta > 0$.

Define the conditional evaluation variance $V_n \triangleq \text{Var}\{\hat{\theta}_r \mid \text{fitted and selected objects}\}$ and assume $nV_n \rightarrow_p \sigma^2 \in (0, \infty)$ and $\hat{V}/V_n \rightarrow_p 1$. Then

$$\frac{\hat{\theta}_r - \theta}{\sqrt{\hat{V}}} \rightsquigarrow N(0, 1),$$

and

$$\hat{\theta}_r \pm z_{1-\alpha/2} \sqrt{\hat{V}}$$

has asymptotic coverage $1 - \alpha$, where $z_{1-\alpha/2}$ is the standard normal quantile. Variance unbiasedness alone does not imply this coverage.

Proposition 3 and Theorem 4 are the exact-ratio benchmarks. The next subsection develops feasible inference when Algorithm 1 uses an estimated ratio $\hat{r}$ that need not equal r_0 .

3.2.2. Estimating the density ratio

This subsection replaces the fixed candidate r above by a fitted ratio $\hat{r}$ in three steps. Propositions 4 and 5 characterize the target r_0 in two ways, as the unique weight that transports OBS averages to RCT averages and as the unique minimizer of a quadratic risk. Definitions 10 to 12 turn these two characterizations into estimators. Proposition 7, Theorem 5, and Corollary 5.1 then track how the error $\hat{r} - r_0$ enters Algorithm 1: it biases the point estimate by an explicit drift but leaves the variance estimator unbiased, and when the drift is small relative to the standard error, the exact-ratio results of Theorems 4 and 3 carry over.

Proposition 4 (Transport characterization). Under either Assumption 3 or Assumption 4, every $h \in L_1(P_R^X)$ satisfies

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}. \quad (32)$$

Conversely, if $\tilde{r} \geq 0$, $\mathbb{E}_O(\tilde{r}) = 1$, and Equation (32) holds with $\tilde{r}$ in place of r_0 for every bounded measurable h , then $\tilde{r} = r_0$ P_O^X -almost surely.

Proposition 5 (Riesz-risk characterization). If $r_0 \in L_2(P_O^X)$, then

$$r_0 \in \arg \min_{a \in L_2(P_O^X)} E_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\}, \quad (33)$$

The minimizer is unique up to P_O^X -almost-sure equality.

Eq. (33) motivates the following learner.

Definition 10 (Empirical Riesz ratio learner). For $S \in \{R, O\}$, let $\mathbb{P}_S^{\text{nuis}}$ denote the empirical average over D_S^{nuis} . For a prespecified function class $\mathcal{A}$, penalty Pen , and $\rho \geq 0$, use a prespecified deterministic rule to break ties among minimizers. Whenever the displayed argmin set is nonempty, choose

$$\hat{r}_{\text{RZ}} \in \arg \min_{a \in \mathcal{A}} \left[\mathbb{P}_O^{\text{nuis}}(a^2) - 2\mathbb{P}_R^{\text{nuis}}(a) + \rho \text{Pen}(a) \right]. \quad (34)$$

The classifier and Riesz routes are alternative base learners; we do not average them.

Equation (34) is Riesz-representer learning of the type used in Auto-DML (Chernozhukov et al., 2021, 2022).

The next lemma specializes Definition 10 to a linear-ridge class and gives explicit conditions for a unique solution.

Lemma 1 (Linear Riesz solution). Let $a_\gamma(x) \triangleq \phi(x)^\top \gamma$, where $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$. Let $\text{Pen}(a_\gamma) \triangleq \|\gamma\|_2^2$. The empirical Gram matrix, namely the average of the feature outer products over the OBS nuisance sample, is $\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\}$. If $\rho > 0$, or if this matrix is nonsingular when $\rho = 0$, then

$$\hat{\gamma}_\rho \triangleq \left[\mathbb{P}_O^{\text{nuis}}\{\phi(X)\phi(X)^\top\} + \rho I_p \right]^{-1} \mathbb{P}_R^{\text{nuis}}\{\phi(X)\}. \quad (35)$$

The classifier learns the ratio indirectly through source prediction.

Definition 11 (Classifier density-ratio estimator). Pool the nuisance-sample covariates and label their source by $S = 1$ for RCT and $S = 0$ for OBS. Define

$$\pi \triangleq P(S = 1), \quad s_0(x) \triangleq P(S = 1 \mid X = x).$$

When $0 < \pi < 1$, Bayes' rule gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)}. \quad (36)$$

For a fitted domain probability $\hat{s}$, let $\hat{\pi}$ denote the effective RCT proportion in classifier

training and define

$$\hat{r}_{\text{CLS}}(x) \triangleq \frac{1 - \hat{\pi}}{\hat{\pi}} \frac{\hat{s}(x)}{1 - \hat{s}(x)}. \quad (37)$$

For an unweighted pooled classifier, $\hat{\pi} \triangleq n_R^{\text{nuis}} / (n_R^{\text{nuis}} + n_O^{\text{nuis}})$. Class weighting or resampling changes the effective training prior, so the prior-odds factor must be corrected accordingly. The odds require $0 < \hat{s}(x) < 1$ where the ratio is used. Probability clipping can stabilize the estimate, but it changes the exact density-ratio target.

The Riesz learner in Definition 10 and the classifier in Definition 11 are the two primary ratio-estimation routes. After either route has been fitted, one may optionally calibrate its positive ratio to selected ATE features.

Definition 12 (Calibrated automatic-balancing ratio). Let $\hat{r}_{\text{base}} > 0$ come from either the classifier or Riesz route. Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^p$ be prespecified with no intercept. Whenever the displayed argmin set is nonempty, choose a deterministically tie-broken minimizer

$$\hat{\xi} \in \arg \min_{\xi \in \mathbb{R}^p} \left[\log \mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\xi^\top \phi(X)} \} - \xi^\top \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \} \right], \quad (38)$$

$$\hat{r}_{\text{BAL}}(x) \triangleq \frac{\hat{r}_{\text{base}}(x) e^{\hat{\xi}^\top \phi(x)}}{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \}}. \quad (39)$$

Normalization already balances the constant, so an intercept would be nonidentified. The minimal ATE feature is $\phi_{\text{ATE}}(X) \triangleq \hat{g}(X)$; a richer finite map can add $\hat{g}^2$, fitted regressions, or a small spline or frozen-neural basis. This exponential tilt follows the calibration principle behind entropy balancing (Hainmueller, 2012).

Proposition 6 (Automatic balance). If the objective in Definition 12 has an interior minimizer, then $\hat{r}_{\text{BAL}} > 0$, $\mathbb{P}_O^{\text{nuis}}(\hat{r}_{\text{BAL}}) = 1$, and

$$\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{BAL}}(X) \phi(X) \} = \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}. \quad (40)$$

These are p finite moment equations, not pointwise balance. They are analogous to covariate-balancing propensity scores, which concern treatment rather than source assignment (Imai and Ratkovic, 2014). Exact feasibility requires the RCT feature mean to lie in the relative interior of the convex hull of the positively weighted OBS feature vectors; otherwise a prespecified approximate-balance rule is required.

We now quantify how the error $\hat{r} - r_0$ propagates through Algorithm 1. The estimated ratio enters the algorithm at exactly two places. In the tuning step it enters the moment $\hat{B}$ of Equation (28), which sets $\hat{\omega}$ through Equation (30). In the evaluation step it enters the OBS score $\psi_O = \hat{\omega} \hat{r} \hat{g}$, which feeds both $\hat{\theta}_{\hat{r}}$ and $\hat{V}$ in Equation (31). The coefficient $\hat{\lambda}$ uses only $\hat{A}$ and $\hat{C}$, so it does not depend on $\hat{r}$. The next proposition treats the two places in turn and then records what they imply for the bias and the variance of $\hat{\theta}_{\hat{r}}$.

Proposition 7 (Effect of the estimated ratio on Algorithm 1). Let $\widehat{B}_{\widehat{r}}$ denote $\widehat{B}$ in Equation (28) evaluated with the estimated ratio $\widehat{r}$, and let $\widehat{B}_{r_0}$ denote the same expression evaluated with the oracle ratio r_0 . Let $\widehat{\theta}_{\widehat{r}}$ and $\widehat{\theta}_{r_0}$ be the estimator in Equation (15) with the same selected coefficients $(\widehat{\lambda}, \widehat{\omega})$ and with $r = \widehat{r}$ and $r = r_0$, respectively. Suppose Assumption 1 holds, $n, N \geq 2$, $\psi_R \in L_2(P_R)$, $\psi_O \in L_2(P_O)$, $\widehat{g} \in L_1(P_R)$, and $r_0\widehat{g}, \widehat{r}\widehat{g} \in L_1(P_O)$.

1. (Tuning step.) On the tuning samples,

$$\widehat{B}_{\widehat{r}} - \widehat{B}_{r_0} = \frac{n}{N} \left[2\widehat{\text{Cov}}_O\{r_0\widehat{g}, (\widehat{r} - r_0)\widehat{g}\} + \widehat{\text{Var}}_O\{(\widehat{r} - r_0)\widehat{g}\} \right]. \quad (41)$$

Hence the ratio error changes $\widehat{\omega} = \widehat{D}/\widehat{B}$ only through its denominator and the threshold t_B in Equation (30), and it leaves $\widehat{\lambda}$ unchanged.

2. (Evaluation step.) On the evaluation samples,

$$\widehat{\theta}_{\widehat{r}} - \widehat{\theta}_{r_0} = \widehat{\omega} \mathbb{P}_{O,N}\{(\widehat{r} - r_0)\widehat{g}\}. \quad (42)$$

3. (Bias and variance.) Conditional on all fitted and selected objects,

$$\mathbb{E}(\widehat{\theta}_{\widehat{r}}) = \theta + \widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}, \quad \mathbb{E}(\widehat{V}) = \text{Var}(\widehat{\theta}_{\widehat{r}}). \quad (43)$$

Item 1 is the coefficient-selection path and item 2 is the evaluation path. Both are small when the OBS sample is much larger than the trial: the difference in item 1 carries the factor n/N , and the difference in item 2 has conditional variance $\widehat{\omega}^2 \text{Var}_O\{(\widehat{r} - r_0)\widehat{g}\}/N$. Item 3 says what remains. The ratio error biases the point estimate by the drift $\widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}$, which is $\widehat{\omega}$ times the transport error of $\widehat{r}$ on the single function $\widehat{g}$, while $\widehat{V}$ stays an unbiased estimator of the conditional variance because it is built from the same scores as $\widehat{\theta}_{\widehat{r}}$. The drift vanishes when $\mathbb{E}_O(\widehat{r}\widehat{g}) = \mathbb{E}_R(\widehat{g})$, which is the population version of the balance that Definition 12 enforces on the nuisance sample with $\phi = \widehat{g}$, and it also vanishes when $\widehat{\omega} = 0$. When the corresponding unconditional moments exist, the unconditional mean of $\widehat{\theta}_{\widehat{r}}$ is θ plus the expected drift, and the unconditional variance additionally contains the between-fit variation of the drift, which $\widehat{V}$ does not capture. Therefore the only new requirement for Wald inference with $\widehat{r}$ is that the drift be small relative to the standard error.

Theorem 5 (Feasible normal approximation with an estimated ratio). Let $\widehat{\theta}_{\widehat{r}}$ and $\widehat{V}$ be the outputs of Algorithm 1 with the estimated ratio $\widehat{r}$, and let $V_n \triangleq \text{Var}(\widehat{\theta}_{\widehat{r}} \mid \text{fitted and selected objects})$ be the conditional variance in item 3 of Proposition 7. Suppose Assumption 1 holds, $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and, for some $\delta > 0$, the centered evaluation scores have conditional $(2+\delta)$ -moments that are uniformly bounded in probability. Suppose also that

$$nV_n \rightarrow_p \sigma^2 \in (0, \infty), \quad \frac{\widehat{\omega}\{\mathbb{E}_O(\widehat{r}\widehat{g}) - \mathbb{E}_R(\widehat{g})\}}{\sqrt{V_n}} = o_p(1). \quad (44)$$

Then

$$\frac{\widehat{V}}{V_n} \rightarrow_p 1, \quad \frac{\widehat{\theta}_{\widehat{r}} - \theta}{\sqrt{\widehat{V}}} \rightsquigarrow N(0, 1), \quad (45)$$

and the interval

$$\hat{\theta}_{\hat{r}} \pm z_{1-\alpha/2} \sqrt{\hat{V}} \quad (46)$$

has asymptotic coverage $1 - \alpha$, where $z_{1-\alpha/2}$ is the standard normal quantile.

Equation (44) states two requirements. The first is that the conditional variance shrinks at the parametric rate with a nondegenerate limit, so that the standard error $\sqrt{V_n}$ is of order $n^{-1/2}$. The second is that the bias caused by the ratio error, the drift of Proposition 7, is negligible relative to that standard error. Equation (45) gives the two conclusions that a Wald interval needs: the variance estimator is consistent for the conditional variance, and the studentized estimator is asymptotically standard normal. Theorem 5 therefore permits $\hat{r} \neq r_0$, and the price is the drift condition. If the drift is not negligible, the interval needs further justification, which may use influence functions, direct bias control, or a different orthogonal construction that accounts for first-stage variation.

Corollary 5.1 (Inheritance of the exact-ratio oracle ordering). Index the three estimators of Theorem 3, built from the same fitted objects $Z_0, \Delta, \hat{g}, \hat{r}$, by $k \in \{\text{RCT}, \text{PPI}^{++}, \text{joint}\}$. Their population-oracle coefficients are

$$\begin{aligned} (\lambda_{\text{RCT}}^*, \omega_{\text{RCT}}^*) &\triangleq (0, 0), & (\lambda_{\text{PPI}^{++}}^*, \omega_{\text{PPI}^{++}}^*) &\triangleq (0, D/B), \\ (\lambda_{\text{joint}}^*, \omega_{\text{joint}}^*) &\triangleq (-C/A, D/B), \end{aligned}$$

and their feasible coefficients $(\hat{\lambda}_k, \hat{\omega}_k)$ are the entries of Equation (30), with $\hat{\lambda}_k \triangleq 0$ for $k \neq \text{joint}$ and $\hat{\omega}_{\text{RCT}} \triangleq 0$. Write $\hat{\theta}_{\hat{r},k} \triangleq \hat{\theta}_{\hat{r}}(\hat{\lambda}_k, \hat{\omega}_k)$, and let σ_k^2 denote the limit σ^2 in Equation (44) for estimator k . Suppose that, for each k , the conditions of Theorem 5 hold, $(\hat{A}, \hat{B}, \hat{C}, \hat{D}) - (A, B, C, D) = o_p(1)$, $(\hat{\lambda}_k, \hat{\omega}_k) - (\lambda_k^*, \omega_k^*) = o_p(1)$,

$$\begin{aligned} \text{Var}_R(\Delta) &= O_p(1), & \text{Var}_R(\hat{g}) + \frac{n}{N} \text{Var}_O(r_0 \hat{g}) &= O_p(1), \\ \mathbb{P}(\hat{A} \leq t_A) + \mathbb{P}(\hat{B} \leq t_B) &\rightarrow 0, \end{aligned} \quad (47)$$

and the evaluation-step difference in item 2 of Proposition 7 is negligible at the $\sqrt{n}$ scale,

$$\hat{\omega}_k^2 \frac{n}{N} \text{Var}_O\{(\hat{r} - r_0) \hat{g}\} = o_p(1). \quad (48)$$

Then, for each k ,

$$\sqrt{n} \left\{ \hat{\theta}_{\hat{r},k} - \hat{\theta}_{r_0}(\lambda_k^*, \omega_k^*) \right\} = o_p(1), \quad (49)$$

so σ_k^2 equals the limit of $\tilde{V}_k$ in Equation (25) and

$$\sigma_{\text{joint}}^2 \leq \sigma_{\text{PPI}^{++}}^2 \leq \sigma_{\text{RCT}}^2. \quad (50)$$

This is not a universal finite-sample dominance statement.

4. Data Fusion for CATE Estimation

In this section, we develop two estimators of the CATE τ in Equation (2) that combine RCT data with potentially confounded OBS data. The RCT identifies the causal target. The OBS outcome regressions and their treatment-effect prediction may improve statistical efficiency, but they need not be causally valid due to unmeasured confounders. The first estimator is the Doubly Robust Fusion (DRF) Learner, based on the AIPW pseudo-outcome of the DR-learner (Kennedy, 2023). The second is the R-Fusion (RF) Learner, based on the R-learner loss (Nie and Wager, 2021) scaled by the known trial propensity variance.

4.1. CATE targets and RCT-valid learning losses

The CATE estimators of this section minimize empirical versions of the following risk.

Definition 13 (CATE risk). Let $\mathcal{Z} \subseteq L_2(P_R^X)$ be a prespecified nonempty class of candidate CATE functions, such as a fixed linear sieve. For $\zeta \in \mathcal{Z}$, define the CATE risk

$$\mathcal{R}(\zeta) \triangleq \mathbb{E}_R\{\zeta(X) - \tau(X)\}^2. \quad (51)$$

Recall the fused score $Z_\lambda = \varphi(V; \hat{\mu}_\lambda)$ of Definition 4. Throughout this section $\lambda \in [0, 1]$, so that $\hat{\mu}_\lambda$ in Definition 4 is a convex combination of the two fitted regressions $\hat{\mu}_R$ and $\hat{\mu}_O$. Under Assumptions 1 and 2, recall that $\mathbb{E}_R(Z_\lambda | X) = \tau(X)$ for every fixed λ . Recall also the OBS prediction $\hat{g} \triangleq \hat{\mu}_{O,1} - \hat{\mu}_{O,0}$ in Equation (14), and write $G_\zeta(X) \triangleq \{\hat{g}(X) - \zeta(X)\}^2$ for the squared distance between the OBS prediction and the candidate CATE function ζ .

We first define the two population risks that the DRF and RF estimators target, and then their empirical versions.

Definition 14 (RCT population risks). For $\lambda \in [0, 1]$ and a candidate CATE function $\zeta \in \mathcal{Z}$, define the doubly robust (DR) risk by

$$\mathcal{L}_{\text{DR}}(\zeta; \lambda) \triangleq \mathbb{E}_R\{Z_\lambda - \zeta(X)\}^2. \quad (52)$$

Let m_λ be the estimate of the RCT marginal outcome regression $\mathbb{E}_R[Y | X]$ built from the same fitted regressions as $\hat{\mu}_\lambda$,

$$\begin{aligned} m_R(x) &\triangleq e(x)\hat{\mu}_{R,1}(x) + \{1 - e(x)\}\hat{\mu}_{R,0}(x), \\ m_O(x) &\triangleq e(x)\hat{\mu}_{O,1}(x) + \{1 - e(x)\}\hat{\mu}_{O,0}(x), \\ m_\lambda &\triangleq (1 - \lambda)m_R + \lambda m_O, \end{aligned}$$

and define the R risk by

$$\mathcal{L}_R(\zeta; \lambda) \triangleq \mathbb{E}_R \left[\frac{[Y - m_\lambda(X) - \{A - e(X)\}\zeta(X)]^2}{e(X)\{1 - e(X)\}} \right]. \quad (53)$$

The DR risk is the mean squared error of regressing the RCT-valid pseudo-outcome Z_λ on X , as in the DR-learner (Kennedy, 2023). It is motivated by the conditional validity $\mathbb{E}_R(Z_\lambda | X) = \tau(X)$

in Equation (6), which makes τ the population regression of Z_λ on X . The R risk is the R-learner loss (Nie and Wager, 2021) with the known trial propensity e , divided by $e(X)\{1 - e(X)\}$, the conditional variance of A given X . Under the unrestricted function class, the minimizer of $\mathcal{L}_R(\zeta; \lambda)$ is again τ , and over $\mathcal{Z}$ it is the $L_2(P_R^X)$ projection of τ ; Proposition 8 states both, so no separate lemma is needed. Both risks are computable from the RCT alone. To use the large OBS covariate sample, the empirical versions below add the same transported correction as the ATE estimator in Equation (15), now applied to the squared distance G_ζ .

Definition 15 (Doubly Robust Fusion and R-Fusion losses). For $S \in \{R, O\}$, let $\mathbb{P}_S^{\text{tune}}$ denote the empirical average over the tuning sample D_S^{tune} . For $\lambda \in [0, 1]$, $\omega \in \mathbb{R}$, a candidate transport weight r , and a candidate CATE function $\zeta \in \mathcal{Z}$, define the Doubly Robust Fusion (DRF) loss by

$$\hat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r) \triangleq \mathbb{P}_R^{\text{tune}}\{Z_\lambda - \zeta(X)\}^2 + \omega \left[\mathbb{P}_O^{\text{tune}}\{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{tune}}\{G_\zeta(X)\} \right]. \quad (54)$$

Define the weight $\chi(A, X) \triangleq \{A - e(X)\}^2 / [e(X)\{1 - e(X)\}]$, which satisfies $\mathbb{E}_R\{\chi(A, X) \mid X\} = 1$ under Assumption 2, and the R-Fusion (RF) loss by

$$\begin{aligned} \hat{L}_{\text{RF}}(\zeta; \lambda, \omega, r) \triangleq & \mathbb{P}_R^{\text{tune}} \left[\frac{[Y - m_\lambda(X) - \{A - e(X)\}\zeta(X)]^2}{e(X)\{1 - e(X)\}} \right] \\ & + \omega \left[\mathbb{P}_O^{\text{tune}}\{r(X)G_\zeta(X)\} - \mathbb{P}_R^{\text{tune}}\{\chi(A, X)G_\zeta(X)\} \right]. \end{aligned} \quad (55)$$

The first term of each loss is the empirical version of the corresponding risk in Definition 14, and the bracket is the OBS correction. Both losses also depend on the fitted objects $\hat{\mu}_R$ and $\hat{\mu}_O$, through Z_λ and m_λ , and on $\hat{g}$, through G_ζ ; these objects are fixed before the tuning samples are used, so the notation displays only the tuning coefficients, and later sections write $\hat{L}_j(\zeta; \lambda, \omega)$ for $j \in \{\text{DRF}, \text{RF}\}$ with r suppressed as well. Any fixed function m may replace m_λ ; we then write $\mathcal{L}_R(\zeta; m)$, $Q_R(m)$, and $\hat{L}_{\text{RF}}(\zeta; m, \omega, r)$, and Section 4.2 uses this form to differentiate in m .

Proposition 8 (Exact targets of the two losses). Suppose Assumptions 2 and 1 hold, and either Assumption 3 holds with $r_0 = 1$ or Assumption 4 holds with $r_0 = dP_R^X/dP_O^X$. Fix $\lambda \in [0, 1]$, $\omega \in \mathbb{R}$, $\zeta \in \mathcal{Z}$, and suppose $Z_\lambda \in L_2(P_R)$, $\tau \in L_2(P_R^X)$, $Y - m_\lambda(X) \in L_2(P_R)$, $G_\zeta \in L_1(P_R)$, and $r_0 G_\zeta \in L_1(P_O)$. Define the noise floors

$$Q_{\text{DR}}(\lambda) \triangleq \mathbb{E}_R\{Z_\lambda - \tau(X)\}^2, \quad Q_R(\lambda) \triangleq \mathbb{E}_R \left[\frac{[Y - m_\lambda(X) - \{A - e(X)\}\tau(X)]^2}{e(X)\{1 - e(X)\}} \right].$$

Then, with $r = r_0$,

$$\begin{aligned} \mathbb{E}\{\hat{L}_{\text{DRF}}(\zeta; \lambda, \omega, r_0)\} &= \mathcal{L}_{\text{DR}}(\zeta; \lambda) = Q_{\text{DR}}(\lambda) + \mathcal{R}(\zeta), \\ \mathbb{E}\{\hat{L}_{\text{RF}}(\zeta; \lambda, \omega, r_0)\} &= \mathcal{L}_R(\zeta; \lambda) = Q_R(\lambda) + \mathcal{R}(\zeta). \end{aligned} \quad (56)$$

Consequently, whenever the minimum is attained, the population minimizers of either expected loss over $\mathcal{Z}$ are the $L_2(P_R^X)$ projections of τ onto $\mathcal{Z}$, for every ω . The same identities hold with any fixed m in place of m_λ .

Proposition 8 shows that both fusion losses target the CATE: minimizing over ζ is unaffected by

the noise floors $Q_{\text{DR}}(\lambda)$ and $Q_R(\lambda)$, which do not depend on ζ . Also, the expected loss is flat in ω for the same reason as in Theorem 1: under exact transport the OBS correction has mean zero.

4.2. Orthogonality of learning equations and nuisance sensitivity

Proposition 8 establishes the population targets. We now ask how each fitted object enters the first-order condition that learns ζ , that is, whether the learning equations are orthogonal to the nuisances in the sense of Definition 6.

Throughout this subsection λ and ω are fixed. The nuisances are the fitted regressions $\hat{\mu}_R$ and $\hat{\mu}_O$ of Section 2.1, which enter the losses through $\hat{\mu}_\lambda$, m_λ , and $\hat{g} = \hat{\mu}_{O,1} - \hat{\mu}_{O,0}$, and the candidate ratio r . Derivatives are the Gateaux derivatives $D_\eta F(\eta)[h]$ of Definition 5.

Definition 16 (Population learning equations). For $j \in \{\text{DRF}, \text{RF}\}$, $\lambda \in [0, 1]$, $\omega \in \mathbb{R}$, a candidate ratio r , $\zeta \in \mathcal{Z}$, and a direction $v \in \mathcal{Z}$, define the learning equation of the expected loss $\mathbb{E}\{\hat{L}_j(\zeta; \lambda, \omega, r)\}$ of Definition 15, as in Definition 6, by

$$\Psi_j(\zeta; \lambda, \omega, r)[v] \triangleq -\frac{1}{2} D_\zeta \mathbb{E}\{\hat{L}_j(\zeta; \lambda, \omega, r)\}[v]. \quad (57)$$

Explicitly,

$$\begin{aligned} \Psi_{\text{DRF}}(\zeta; \lambda, \omega, r)[v] &= \mathbb{E}_R[v(X)\{Z_\lambda - \zeta(X)\}] \\ &\quad + \omega [\mathbb{E}_O\{r(X)v(X)[\hat{g}(X) - \zeta(X)]\} - \mathbb{E}_R\{v(X)[\hat{g}(X) - \zeta(X)]\}], \end{aligned} \quad (58)$$

$$\begin{aligned} \Psi_{\text{RF}}(\zeta; \lambda, \omega, r)[v] &= \mathbb{E}_R \left[v(X) \frac{A - e(X)}{e(X)\{1 - e(X)\}} \{Y - m_\lambda(X) - [A - e(X)]\zeta(X)\} \right] \\ &\quad + \omega [\mathbb{E}_O\{r(X)v(X)[\hat{g}(X) - \zeta(X)]\} - \mathbb{E}_R\{\chi(A, X)v(X)[\hat{g}(X) - \zeta(X)]\}]. \end{aligned} \quad (59)$$

The expected loss is quadratic in ζ and, at $r = r_0$, convex by Proposition 8, so Definition 6 applies: the population minimizer over $\mathcal{Z}$ solves $\Psi_j(\zeta; \lambda, \omega, r_0)[v] = 0$ for every $v \in \mathcal{Z}$, and the tuning-sample optimization solves the empirical version of the same equation. The factor $-\frac{1}{2}$ removes the constant produced by differentiating a square, so that Ψ_j is the cross-moment of the direction v with the residual of the loss, plus the corresponding OBS correction. A nuisance error moves the fitted function only through its effect on Ψ_j , which is what the next proposition measures.

At $r = r_0$, the OBS bracket of each loss has expectation zero for every ζ , which is the last step in the proof of Proposition 8, so it contributes nothing to the population learning equations:

$$\begin{aligned} \Psi_{\text{DRF}}(\zeta; \lambda, \omega, r_0)[v] &= \mathbb{E}_R[v(X)\{Z_\lambda - \zeta(X)\}], \\ \Psi_{\text{RF}}(\zeta; \lambda, \omega, r_0)[v] &= \mathbb{E}_R \left[v(X) \frac{A - e(X)}{e(X)\{1 - e(X)\}} \{Y - m_\lambda(X) - [A - e(X)]\zeta(X)\} \right]. \end{aligned}$$

Exact transport thus removes the OBS correction from the population learning equations, while the correction remains in the empirical optimization problem.

Proposition 9 (Directional derivatives of the learning equations). Suppose Assumptions 2 and 1 hold, either Assumption 3 or Assumption 4 holds, and differentiation and expectation can be interchanged along the displayed paths. Fix $j \in \{\text{DRF}, \text{RF}\}$ and $v \in \mathcal{Z}$. For a pair of directions $h = (h_0, h_1)$, let $D_{\hat{\mu}_R} \Psi_j[h, v]$ and $D_{\hat{\mu}_O} \Psi_j[h, v]$ denote the derivatives of $\Psi_j(\zeta; \lambda, \omega, r)[v]$ along $\hat{\mu}_R + \varepsilon h$ and along $\hat{\mu}_O + \varepsilon h$, and for a direction h_r let $D_r \Psi_j[h_r, v]$ denote the derivative along $r + \varepsilon h_r$.

1. (RCT regression.) $D_{\hat{\mu}_R} \Psi_j[h, v] = 0$.

2. (OBS regression.) With $u \triangleq h_1 - h_0$ the induced direction of $\hat{g}$,

$$D_{\hat{\mu}_O} \Psi_j[h, v] = \omega \mathbb{E}_O[\{r(X) - r_0(X)\}u(X)v(X)], \quad (60)$$

which vanishes when $r = r_0$ or $\omega = 0$, but not for an approximate ratio.

3. (Density ratio.)

$$D_r \Psi_j[h_r, v] = \omega \mathbb{E}_O[h_r(X)v(X)\{\hat{g}(X) - \zeta(X)\}]. \quad (61)$$

The directions that preserve the normalization $\mathbb{E}_O(r) = 1$ are those with $\mathbb{E}_O(h_r) = 0$. If $\omega \neq 0$ and $v(\hat{g} - \zeta) \in L_2(P_O^X)$, then $D_r \Psi_j[h_r, v] = 0$ for every such $h_r \in L_2(P_O^X)$ if and only if $v(X)\{\hat{g}(X) - \zeta(X)\}$ is P_O^X -almost surely constant.

Item 1 is the classical orthogonality of the AIPW score and of the R loss to their RCT outcome nuisances under a known propensity (Chernozhukov et al., 2018; Kennedy, 2023; Nie and Wager, 2021). Item 2 says that the OBS regression is a first-order nuisance only through the transported correction: its effect through $\hat{\mu}_\lambda$ and m_λ vanishes for the same reason as in item 1, and its effect through $\hat{g}$ vanishes exactly when the ratio is exact. Item 3 says that the learning equations are not orthogonal to the density ratio: a first-order error in $\hat{r}$ moves the learning equation by Equation (61), and calibrating $\hat{r}$ to finitely many moments, as in Definition 12, constrains h_r only on a finite-dimensional subspace and does not change this conclusion.

4.3. Fitting the learners and choosing the fusion coefficients

The DRF and RF learners have three inputs that the data must supply: the fitted CATE function for given coefficients, the coefficients (λ, ω) , and the ratio r . Algorithm 2 states the procedure; the rest of this subsection defines its objects and analyzes its output, treating the three inputs in turn.

Algorithm 2: Honest fitting and selection of the CATE fusion learner

Input: Nuisance, tuning, and evaluation samples; the class $\mathcal{Z}$; a prespecified finite grid $\mathcal{G}_j \subseteq [0, 1]^2$ containing $(0, 0)$; a density-ratio estimator.

- 1 Fit $\hat{\mu}_R$, $\hat{\mu}_O$, $\hat{g}$, and r on the nuisance samples.
- 2 For every $(\lambda, \omega) \in \mathcal{G}_j$, fit $\hat{\zeta}_j(\cdot; \lambda, \omega, r)$ on the tuning samples by minimizing $\hat{L}_j(\zeta; \lambda, \omega, r)$ of Definition 15 over $\zeta \in \mathcal{Z}$.
- 3 For every $(\lambda, \omega) \in \mathcal{G}_j$, compute on the evaluation samples, with $G \triangleq \{\hat{g}(X) - \hat{\zeta}_j(X; \lambda, \omega, r)\}^2$, the validation score $\hat{C}_j(\lambda, \omega) \triangleq \mathbb{P}_{R,n}[\{Z_0 - \hat{\zeta}_j(X; \lambda, \omega, r)\}^2 - \omega G] + \mathbb{P}_{O,N}\{\omega r(X)G\}$.
- 4 Select $(\hat{\lambda}_j, \hat{\omega}_j)$ by minimizing $\hat{C}_j$ over $\mathcal{G}_j$, with deterministic tie-breaking.

Output: $(\hat{\lambda}_j, \hat{\omega}_j)$ and $\hat{\zeta}_j(\cdot; \hat{\lambda}_j, \hat{\omega}_j, r)$.

Definition 17 (Fitted fusion learner). Fix $j \in \{\text{DRF}, \text{RF}\}$, a prespecified compact set $\Omega \subseteq [0, 1]$ containing 0, coefficients $(\lambda, \omega) \in [0, 1] \times \Omega$, and a candidate ratio r . The fitted fusion learner is

$$\hat{\zeta}_j(\cdot; \lambda, \omega, r) \in \arg \min_{\zeta \in \mathcal{Z}} \hat{L}_j(\zeta; \lambda, \omega, r), \quad (62)$$

with ties broken by a prespecified deterministic rule, and its CATE risk is $\mathcal{R}\{\hat{\zeta}_j(\cdot; \lambda, \omega, r)\}$ in Equation (51), evaluated conditionally on all fitted objects.

Two abbreviations are used below: $\hat{\zeta}_j(\cdot; \lambda, \omega)$ is the learner fitted with the ratio r in use, and $\hat{\zeta}_j(\cdot; 0, 0)$ is the RCT-only learner, which uses neither ω nor r .

Step 2 of Algorithm 2 is a convex least-squares problem, because the losses of Definition 15 are quadratic in ζ with a nonnegative second-order part; the expansion is recorded in Appendix B.1.

In DRF, λ changes Z_λ ; in RF, it changes m_λ ; in both, ω scales the OBS correction and r weights it. The fitted function therefore depends on (λ, ω, r) , even though by Proposition 8 the expected loss at a fixed ζ does not depend on ω at $r = r_0$. The coefficients that minimize the CATE risk of the fitted function cannot be computed, because τ is unknown, so the coefficients are chosen by comparing the fitted candidates on the evaluation samples, as in steps 3 and 4 of Algorithm 2.

Definition 18 (Validation score on a finite grid). Let $\mathcal{G}_j \subseteq [0, 1] \times \Omega$ be a prespecified finite grid containing $(0, 0)$, and let $\hat{\zeta}_j(\cdot; \lambda, \omega)$, $(\lambda, \omega) \in \mathcal{G}_j$, be the candidates fitted in step 2 of Algorithm 2 with the ratio r . For each grid pair, define on the evaluation samples $G_{j,\lambda,\omega}(X) \triangleq \{\hat{g}(X) - \hat{\zeta}_j(X; \lambda, \omega)\}^2$ and the validation score of step 3,

$$\hat{C}_j(\lambda, \omega) \triangleq \mathbb{P}_{R,n}[\{Z_0 - \hat{\zeta}_j(X; \lambda, \omega)\}^2 - \omega G_{j,\lambda,\omega}(X)] + \mathbb{P}_{O,N}\{\omega r(X)G_{j,\lambda,\omega}(X)\}. \quad (63)$$

Step 4 selects $(\hat{\lambda}_j, \hat{\omega}_j) \in \arg \min_{(\lambda, \omega) \in \mathcal{G}_j} \hat{C}_j(\lambda, \omega)$, with deterministic tie-breaking.

The score uses the same pseudo-outcome Z_0 for every candidate, so that candidates differ only through their fitted functions, and it adds the transported correction $\omega\{\mathbb{P}_{O,N}(rG_{j,\lambda,\omega}) - \mathbb{P}_{R,n}(G_{j,\lambda,\omega})\}$, whose conditional mean is zero when $r = r_0$, exactly as the control variate in Equation (15).

Theorem 6 (Honest validation of the fusion candidates). Suppose Assumptions 1 and 2 hold, and either Assumption 3 holds with $r_0 = 1$ or Assumption 4 holds with $r_0 = dP_R^X/dP_O^X$. Let r and the candidates be as in Definition 18, suppose $|Z_0|$, $|\hat{g}|$, and $|\hat{\zeta}_j(\cdot; \lambda, \omega)|$ are bounded by B and $0 \leq r \leq \bar{r}$, let $M \triangleq |\mathcal{G}_j|$, and for $\alpha \in (0, 1)$ define

$$\varepsilon_{\text{eval}}(\alpha) \triangleq 2B^2 \sqrt{2 \log(4M/\alpha)} \left(\frac{2}{\sqrt{n}} + \frac{\bar{r}}{\sqrt{N}} \right). \quad (64)$$

Conditional on all fitted objects, the following hold.

1. (What the score measures.) For every $(\lambda, \omega) \in \mathcal{G}_j$,

$$\mathbb{E}\{\hat{C}_j(\lambda, \omega)\} = Q_{\text{DR}}(0) + \mathcal{R}\{\hat{\zeta}_j(\cdot; \lambda, \omega)\} + \omega \mathbb{E}_O[\{r(X) - r_0(X)\}G_{j,\lambda,\omega}(X)]. \quad (65)$$

The first term is the same for every candidate, and the last term, the transport error of r , is zero when $r = r_0$ and for the RCT-only candidate.

2. (How accurate the score is.) With probability at least $1 - \alpha$ over the evaluation samples, $|\hat{C}_j(\lambda, \omega) - \mathbb{E}\{\hat{C}_j(\lambda, \omega)\}| \leq \varepsilon_{\text{eval}}(\alpha)$ for every $(\lambda, \omega) \in \mathcal{G}_j$.
3. (What the selection achieves.) On the same event,

$$\begin{aligned} \mathcal{R}\{\hat{\zeta}_j(\cdot; \hat{\lambda}_j, \hat{\omega}_j)\} &\leq \min_{(\lambda, \omega) \in \mathcal{G}_j} \mathcal{R}\{\hat{\zeta}_j(\cdot; \lambda, \omega)\} + 2\varepsilon_{\text{eval}}(\alpha) \\ &\quad + 2 \max_{(\lambda, \omega) \in \mathcal{G}_j} |\omega \mathbb{E}_O[\{r(X) - r_0(X)\}G_{j,\lambda,\omega}(X)]|, \end{aligned} \quad (66)$$

and the minimum is at most $\mathcal{R}\{\hat{\zeta}_j(\cdot; 0, 0)\}$ because $(0, 0) \in \mathcal{G}_j$.

In words, the score orders the candidates by their CATE risk up to a constant that is common to all of them; the order is correct up to $\varepsilon_{\text{eval}}(\alpha)$, which decreases like $n^{-1/2}$ and $N^{-1/2}$ and grows only like $\sqrt{\log M}$ in the number of candidates; so the selected learner is within $2\varepsilon_{\text{eval}}(\alpha)$ of the best candidate, and in particular of the RCT-only learner. An estimated ratio from Section 3.2.2 adds twice its largest transport error on the functions $G_{j,\lambda,\omega}$; under Assumption 3, $r_0 = 1$ and no ratio is estimated.

4.4. Linear-sieve and additive-spline specialization

This specialization has two roles. It computes every fixed (λ, ω) candidate used in Section 4.3, and it turns the abstract two-source modulus into explicit covariance and approximation terms.

Definition 19 (Linear-sieve fusion estimator). Let $b(x) \in \mathbb{R}^p$ be fixed or honestly trained and define $\zeta_\beta(x) \triangleq b(x)^\top \beta$. Let $P = P^\top \succeq 0$ and $\rho \geq 0$. For $j \in \{\text{DRF}, \text{RF}\}$, define

$$\begin{aligned} \hat{J}_j(\beta; \lambda, \omega) &\triangleq \hat{L}_j(\zeta_\beta; \lambda, \omega) + \rho \beta^\top P \beta, \\ \hat{\beta}_{\lambda, \omega}^j &\in \arg \min_{\beta \in \mathbb{R}^p} \hat{J}_j(\beta; \lambda, \omega), \\ \hat{\zeta}_j(x; \lambda, \omega) &\triangleq b(x)^\top \hat{\beta}_{\lambda, \omega}^j, \end{aligned}$$

using deterministic tie-breaking whenever needed.

For DRF, define

$$\begin{aligned}\hat{\Gamma}_R &\triangleq \mathbb{P}_R^{\text{tune}}(bb^\top), & \hat{\Gamma}_{O,r} &\triangleq \mathbb{P}_O^{\text{tune}}(r_0bb^\top), \\ \hat{h}_\lambda &\triangleq \mathbb{P}_R^{\text{tune}}(bZ_\lambda), & \hat{d}_{\text{OBS}} &\triangleq \mathbb{P}_O^{\text{tune}}(r_0b\hat{g}) - \mathbb{P}_R^{\text{tune}}(b\hat{g}).\end{aligned}$$

For RF, let

$$\begin{aligned}s_{m_\lambda} &\triangleq \frac{(A-e)(Y-m_\lambda)}{e(1-e)}, & \hat{\Gamma}_{R,\chi} &\triangleq \mathbb{P}_R^{\text{tune}}(\chi bb^\top), \\ \hat{h}_{m_\lambda} &\triangleq \mathbb{P}_R^{\text{tune}}(bs_{m_\lambda}), & \hat{d}_{\text{OBS},\chi} &\triangleq \mathbb{P}_O^{\text{tune}}(r_0b\hat{g}) - \mathbb{P}_R^{\text{tune}}(\chi b\hat{g}).\end{aligned}$$

Proposition 10 (Normal equations and positive semidefiniteness). For DRF and RF, respectively, define

$$\begin{aligned}\hat{H}_{\text{DRF}}(\omega) &\triangleq (1-\omega)\hat{\Gamma}_R + \omega\hat{\Gamma}_{O,r}, & \hat{h}_{\text{DRF}}(\lambda, \omega) &\triangleq \hat{h}_\lambda + \omega\hat{d}_{\text{OBS}}, \\ \hat{H}_{\text{RF}}(\omega) &\triangleq (1-\omega)\hat{\Gamma}_{R,\chi} + \omega\hat{\Gamma}_{O,r}, & \hat{h}_{\text{RF}}(\lambda, \omega) &\triangleq \hat{h}_{m_\lambda} + \omega\hat{d}_{\text{OBS},\chi}.\end{aligned}$$

The normal equation is

$$\{\hat{H}_j(\omega) + \rho P\}\hat{\beta}_{\lambda,\omega}^j = \hat{h}_j(\lambda, \omega).$$

When the matrix is invertible,

$$\hat{\beta}_{\lambda,\omega}^{\text{DRF}} = \{\hat{H}_{\text{DRF}}(\omega) + \rho P\}^{-1}\hat{h}_{\text{DRF}}(\lambda, \omega), \quad (67)$$

$$\hat{\beta}_{\lambda,\omega}^{\text{RF}} = \{\hat{H}_{\text{RF}}(\omega) + \rho P\}^{-1}\hat{h}_{\text{RF}}(\lambda, \omega). \quad (68)$$

If $0 \leq \omega \leq 1$, the ratio weights are nonnegative, and $P \succeq 0$, both penalized Hessians are positive semidefinite. The signed source representation alone does not imply invertibility. A fit using $\hat{r}$ requires nonnegative weights, a direct Hessian check, or a prespecified positive-semidefinite projection.

For fixed ω , the Hessian does not depend on λ , while the right-hand side is affine. This yields an efficient candidate-generation device, not a replacement for final honest validation.

Proposition 11 (Fixed-scale profiling for candidate generation). Suppose $\hat{H}_j(\omega) + \rho P$ is invertible and write

$$\hat{h}_j(\lambda, \omega) \triangleq \hat{h}_{j,0}(\omega) + \lambda\hat{h}_{j,1}(\omega).$$

Then

$$\hat{\beta}_{\lambda,\omega}^j = \hat{\beta}_{0,\omega}^j + \lambda\hat{\beta}_\omega^j, \quad \hat{\beta}_\omega^j \triangleq \{\hat{H}_j(\omega) + \rho P\}^{-1}\hat{h}_{j,1}(\omega). \quad (69)$$

On a separate tuning sample, the corresponding quadratic criterion has the form

$$\hat{C}_j^{\text{tune}}(\lambda, \omega) = c_0(\omega) + 2c_1(\omega)\lambda + c_2(\omega)\lambda^2. \quad (70)$$

For a prespecified finite ω -grid, checking the admissible stationary point and the endpoints of $[0, 1]$ generates finitely many candidates. Those candidates must be fixed before the final validation used in Theorem 6. Continuous profiling on the final evaluation sample requires a separate uniform-convergence argument.

We now separate sieve approximation from estimation. Define

$$\begin{aligned} \Gamma_R &\triangleq \mathbb{E}_R(bb^\top), & \gamma_R &\triangleq \mathbb{E}_R(b\tau), \\ \beta_p &\triangleq \Gamma_R^{-1}\gamma_R, & \zeta_p &\triangleq b^\top\beta_p, & a_p^2 &\triangleq \mathbb{E}_R(\tau - \zeta_p)^2, \end{aligned}$$

and assume $\Gamma_R \succ 0$. For a fixed candidate, let $\hat{H}_j \triangleq \hat{H}_j(\omega)$, $\hat{h}_j \triangleq \hat{h}_j(\lambda, \omega)$, and define

$$\hat{u}_j \triangleq \hat{h}_j - \hat{H}_j\beta_p.$$

Theorem 7 (Exact random-Hessian sieve identity). For any realized tuning samples and fixed fitted nuisances, if $\hat{H}_j + \rho P$ is invertible, then

$$\hat{\beta}_j - \beta_p = (\hat{H}_j + \rho P)^{-1}\{\hat{u}_j - \rho P\beta_p\}, \quad (71)$$

$$\mathcal{R}(\hat{\zeta}_j) = a_p^2 + (\hat{\beta}_j - \beta_p)^\top \Gamma_R (\hat{\beta}_j - \beta_p). \quad (72)$$

The identity is exact, but it is not a same-sample bias-variance decomposition because the random inverse Hessian and score are dependent.

For the concentration result, define the raw source contributions explicitly:

$$\begin{aligned} \psi_{R,\text{DRF}} &\triangleq b\{Z_\lambda - \omega\hat{g} - (1 - \omega)\zeta_p\}, & \psi_{O,\text{DRF}} &\triangleq \omega r_0 b(\hat{g} - \zeta_p), \\ \psi_{R,\text{RF}} &\triangleq b[s_{m_\lambda} - \chi\{\omega\hat{g} + (1 - \omega)\zeta_p\}], & \psi_{O,\text{RF}} &\triangleq \omega r_0 b(\hat{g} - \zeta_p). \end{aligned}$$

Under the exact-target conditions, $\mathbb{E}_R\psi_{R,j} + \mathbb{E}_O\psi_{O,j} = 0$; the two source means need not vanish separately. Let

$$\Sigma_{R,j} \triangleq \text{Var}_R(\psi_{R,j}), \quad \Sigma_{O,j} \triangleq \text{Var}_O(\psi_{O,j}).$$

For $S \in \{R, O\}$, define

$$\nu_{S,j} \triangleq \|\Gamma_R^{-1/2}\Sigma_{S,j}\Gamma_R^{-1/2}\|_{\text{op}}.$$

Theorem 8 (Fixed-candidate covariance-sensitive bound). Assume RCT consistency and conditional randomization, known strict-overlap e , exact r_0 , honestly fixed nuisances and basis, fixed $(j, \lambda, \omega, p, \rho)$, and independent tuning samples of sizes n_f, N_f . Assume the centered source vectors are sub-Gaussian with the displayed covariance proxies and the weighted

designs satisfy, with failure probability at most δ_H ,

$$\mathcal{E}_H \triangleq \{\|\Gamma_R^{-1/2}(\hat{H}_j - \Gamma_R)\Gamma_R^{-1/2}\|_{\text{op}} \leq \eta\}, \quad 0 \leq \eta < 1.$$

Then, with probability at least $1 - \delta_H - \delta_S$,

$$\begin{aligned} \mathcal{R}(\hat{\zeta}_j) \leq & a_p^2 + \frac{C_{\text{sg}}}{(1-\eta)^2} \left[\frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j}) + \nu_{R,j} \log(4/\delta_S)}{n_f} \right. \\ & \left. + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j}) + \nu_{O,j} \log(4/\delta_S)}{N_f} \right] + \frac{C_\rho \rho^2}{(1-\eta)^2} \|\Gamma_R^{-1/2} P \beta_p\|_2^2, \end{aligned} \quad (73)$$

where $\nu_{S,j}$ bounds the corresponding whitened covariance operator norm and C_{sg}, C_ρ are universal constants under the stated sub-Gaussian convention. Independence makes the source covariance proxies add; the bound has no artificial $1/\sqrt{n_f N_f}$ cross term.

The preceding theorem is a safe finite-sample upper bound. A trace expression for the leading risk requires a separate fixed-dimensional asymptotic result.

Proposition 12 (Fixed-dimensional leading expected risk). Assume RCT consistency and conditional randomization, known strict-overlap e , honest fixed nuisances and basis, and either a common covariate marginal with $r_0 = 1$ or $P_R^X \ll P_O^X$ with exact $r_0 = dP_R^X/dP_O^X$. Fix (j, λ, ω, p) , and suppose the RCT and OBS tuning samples are independent with $n_f, N_f \rightarrow \infty$ at a stable sample-size ratio. Assume finite $(2 + \delta)$ -moments, $\Gamma_R \succ 0$, Gram convergence, and first-order negligible ridge. Define the linearization remainder

$$R_{n_f, N_f} \triangleq \hat{\beta}_j - \beta_p - \Gamma_R^{-1} [(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}],$$

and assume explicitly that

$$\mathbb{E}\|R_{n_f, N_f}\|_{\Gamma_R}^2 = o(n_f^{-1} + N_f^{-1}), \quad \|a\|_{\Gamma_R}^2 \triangleq a^\top \Gamma_R a.$$

Then

$$\hat{\beta}_j - \beta_p = \Gamma_R^{-1} [(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}] + R_{n_f, N_f},$$

and

$$\mathbb{E}\{\mathcal{R}(\hat{\zeta}_j)\} - a_p^2 = \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j})}{n_f} + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j})}{N_f} + o(n_f^{-1} + N_f^{-1}). \quad (74)$$

This is an asymptotic expected leading-risk expansion, not an exact finite-sample identity.

Corollary 8.1 (First-order RCT versus fusion comparison). Assume RCT consistency and conditional randomization, known strict-overlap e , honest fixed nuisances and basis, and either a common covariate marginal with $r_0 = 1$ or $P_R^X \ll P_O^X$ with exact $r_0 = dP_R^X/dP_O^X$. Assume independent RCT and OBS tuning samples with $n_f, N_f \rightarrow \infty$ at a stable sample-size ratio. Consider a fixed fusion rule (λ, ω) and the strict RCT-only comparator $(\lambda, \omega) = (0, 0)$.

The two rules use the same basis, penalty, RCT nuisance, and sample split. Assume finite $(2+\delta)$ -moments, $\Gamma_R \succ 0$, Gram convergence, first-order negligible ridge, and for each rule the mean-square linearization remainder is $o(n_f^{-1} + N_f^{-1})$. Their difference in first-order expected excess risk is

$$\mathcal{V}_{\text{FUS}}^{(1)} - \mathcal{V}_{\text{RCT}}^{(1)} \triangleq \frac{\text{tr}[\Gamma_R^{-1}\{\Sigma_{R,j}(\lambda, \omega) - \Sigma_{R,j}(0, 0)\}]}{n_f} + \frac{\text{tr}[\Gamma_R^{-1}\Sigma_{O,j}(\lambda, \omega)]}{N_f}. \quad (75)$$

The fixed fusion rule improves on the strict RCT-only comparator at first order if and only if the right-hand side of Equation (75) is negative. Equality does not order the higher-order remainders, and this comparison gives no universal finite-sample dominance.

An estimated ratio can perturb both the normal-equation right-hand side and its Hessian. Define

$$\hat{\Delta}_{h,j} \triangleq \omega \mathbb{P}_O^{\text{tune}}[(\hat{r} - r_0)b\hat{g}], \quad (76)$$

$$\hat{\Delta}_{H,j} \triangleq \omega \mathbb{P}_O^{\text{tune}}[(\hat{r} - r_0)bb^\top]. \quad (77)$$

Proposition 13 (Estimated-ratio perturbation of a fitted sieve). Condition on an independent ratio-training sample and use the same candidate specification with r_0 and $\hat{r}$. If both penalized Hessians are invertible, then

$$\hat{\beta}_j^{\hat{r}} - \hat{\beta}_j^{r_0} = \{\hat{H}_j^{\hat{r}} + \rho P\}^{-1}\{\hat{\Delta}_{h,j} - \hat{\Delta}_{H,j}\hat{\beta}_j^{r_0}\}.$$

Thus ratio normalization alone controls neither the score nor Hessian perturbation. This proposition is the sieve form of item 3 of Proposition 9; it is not an orthogonality result.

Finally, the additive specialization exposes how the two sample sizes enter the rate through method-specific variance constants rather than a generic effective sample size.

Corollary 8.2 (Operational additive-spline rate). Suppose $\tau(x) = \sum_{\ell=1}^d \tau_\ell(x_\ell)$, with fixed d and smoothness $s > 0$. Let K denote the spline dimension and suppose $a_K^2 = O(K^{-2s})$. Assume RCT consistency and randomization, known strict-overlap e , exact r_0 unless an explicit fitting-drift term is retained, honest fixed nuisances and basis, independent tuning samples, and uniformly bounded eigenvalues of Γ_R . Assume the centered source scores and weighted designs have uniformly bounded sub-Gaussian norms, $K + \log(1/\delta) = o\{\min(n_f, N_f)\}$, and the Gram event holds with probability tending to one. Define the per-dimension variance scale

$$c_{S,j}(K; \lambda, \omega) \triangleq K^{-1} \text{tr}\{\Gamma_R(K)^{-1} \Sigma_{S,j}^{(K)}(\lambda, \omega)\}, \quad S \in \{R, O\},$$

$$v_j(K; \lambda, \omega) \triangleq \frac{c_{R,j}(K; \lambda, \omega)}{n_f} + \frac{c_{O,j}(K; \lambda, \omega)}{N_f},$$

and suppress λ from $c_{O,j}$ only when the OBS score contribution is invariant to it. If the fitting ratio drift and ridge bias are $B_{r,j}^{\text{tune}}$ and $B_{\rho,j}$, then

$$\mathcal{R}(\hat{\zeta}_j) = O_p\{K^{-2s} + K v_j(K; \lambda, \omega) + B_{r,j}^{\text{tune}} + B_{\rho,j}\}. \quad (78)$$

For a prespecified admissible dimension set $\mathcal{K}_n$, define the noncircular oracle dimension by

$$K_j^*(\lambda, \omega) \in \arg \min_{K \in \mathcal{K}_n} \{K^{-2s} + K v_j(K; \lambda, \omega)\}.$$

If, along the admissible sequence and uniformly over the coefficient candidates under consideration, $c_{S,j}(K; \lambda, \omega) \rightarrow c_{S,j}^\infty(\lambda, \omega)$, where the limiting constants are uniformly bounded away from zero and infinity, define

$$v_j^\infty(\lambda, \omega) \triangleq \frac{c_{R,j}^\infty(\lambda, \omega)}{n_f} + \frac{c_{O,j}^\infty(\lambda, \omega)}{N_f}.$$

Then a rate-optimal admissible choice satisfies $K \asymp \{v_j^\infty(\lambda, \omega)\}^{-1/(2s+1)}$, and the first two terms are of order $\{v_j^\infty(\lambda, \omega)\}^{2s/(2s+1)}$. For a validation-selected learner, the validation radius and ratio drift of Theorem 6 must also be added. The harmonic size $(1/n_f + 1/N_f)^{-1}$ is only a rough simplification when the two variance constants have comparable, uniformly bounded order.

4.5. Implementation boundaries and open theory

A fixed or honestly trained frozen neural representation can replace b , conditional on that representation. End-to-end neural fitting is nonconvex and is not covered. Estimated-ratio guarantees require uniform score, Hessian, and validation-loss control. Cross-fitting can recover data efficiency, but overlapping outer training sets do not justify a naive sum of fold variances. Neither target preservation nor candidate inclusion gives universal finite-sample improvement for estimated (λ, ω) .

5. Experiments

5.1. Completed ATE benchmarks

The completed experiments evaluate ATE estimation only. All nested benchmarks use 20 Monte Carlo repetitions per cell, an RCT-only AIPW baseline, a λ -only outcome-model fusion estimator, an ω -only PPI correction estimator, and the joint estimator. The code retains two stale internal names: `haipw_only` denotes the λ -only estimator and `ppi_power_tuned_only` denotes the ω -only estimator. The first label does not identify the method with the separate HAIPW literature.

The fixed-OBS-budget benchmark contains 16 design cells: three arbitrary nonlinear SCMs and one STAR real- X construction, each under a common covariate marginal and covariate shift and $n_R \in \{20, 30\}$, with $N_O = 5000$. The joint estimator has the smallest RMSE in 11 cells, and the λ -only estimator is best in the remaining five. The joint estimator has lower RMSE than AIPW in all 16 cells, with mean RMSE ratio 0.648 and mean empirical-variance ratio 0.439 relative to AIPW. The λ -only estimator also beats AIPW in all 16 cells, with mean RMSE ratio 0.679. The ω -only estimator is not best in any cell and has mean RMSE ratio 1.005.

The proportional-budget benchmark contains 32 cells with $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

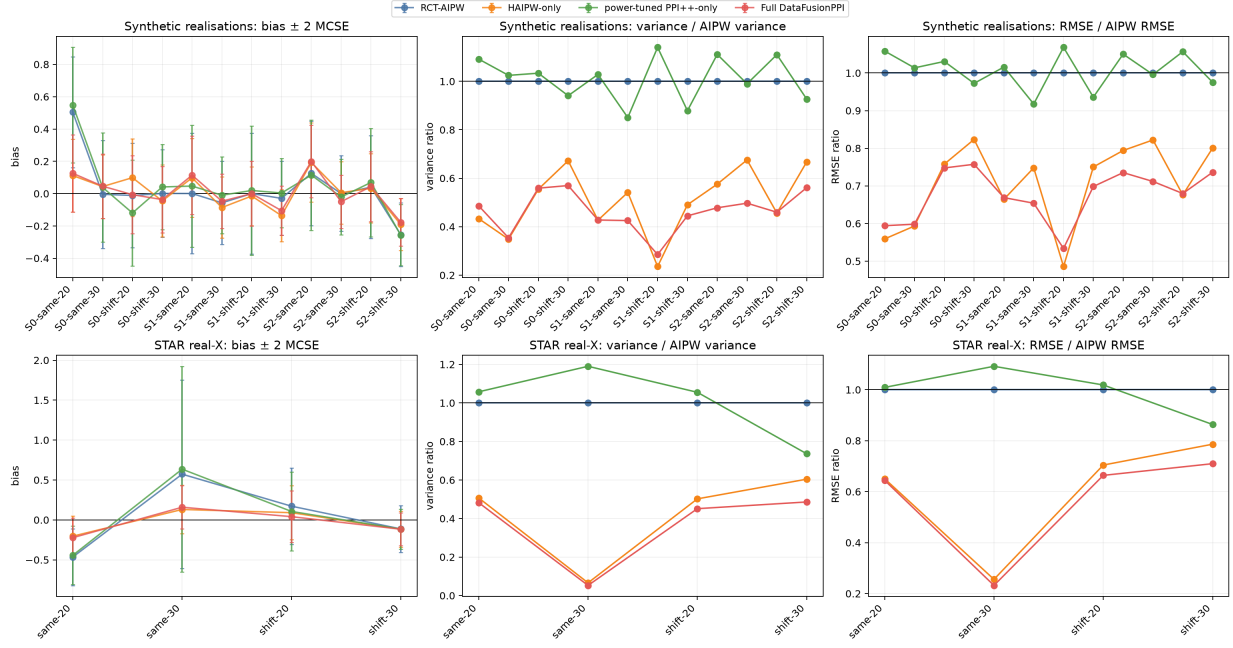


Figure 1: Fixed-OBS-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30\}$ and $N_O = 5000$.

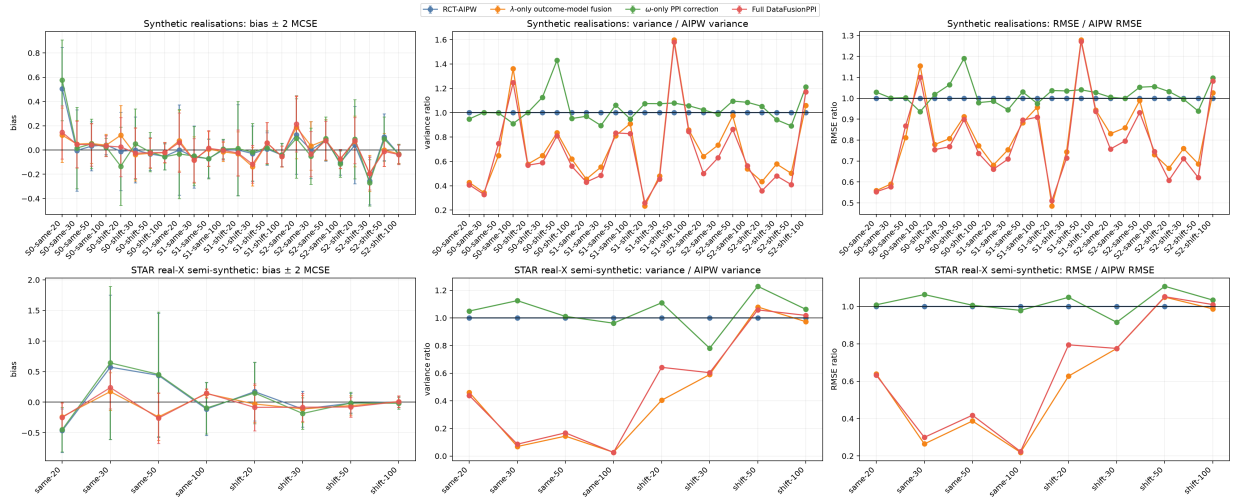


Figure 2: Proportional-budget ATE benchmark. The composite artifact reports multiple method and regime panels side by side for $n_R \in \{20, 30, 50, 100\}$ and $N_O = 100n_R$.

The joint estimator has the smallest RMSE in 18 cells, the λ -only estimator in 10, the ω -only estimator in one, and AIPW in three. The joint estimator has lower RMSE than AIPW in 27 of 32 cells, with mean RMSE ratio 0.760 and mean empirical-variance ratio 0.625. The λ -only estimator beats AIPW in 28 cells and has mean RMSE ratio 0.769. The ω -only estimator beats AIPW in 10 cells and has mean RMSE ratio 1.020. These counts show that joint tuning is often useful but is not uniformly best after coefficient estimation.

The STAR benchmark uses only real kindergarten covariates from Project STAR. Treatment assignments, latent confounding, potential outcomes, and effect heterogeneity are generated synthetically. It is therefore a real- X , semi-synthetic benchmark rather than an analysis of STAR’s original intervention or outcomes. The planned WHI CT+OS analysis has not been run because participant-level access, a semantic crosswalk, and a survival or censoring extension are still required.

5.2. Planned CATE evaluation and interpretation

CATE experiments are planned. The primary design will compare the proved AIPW-pseudo-outcome F-Learner with the prospective R-loss extension using additive cubic B-splines, frozen neural bases, and linear sieves under a common covariate marginal and covariate shift. Each study will compare trial-only, λ -only, ω -only, and joint tuning using held-out RCT-population CATE risk. Classifier-only, balancing-only, hybrid, and oracle transport weights will be separated. No CATE result is claimed from Figures 1 and 2.

The empirical conclusion is deliberately limited. The two-channel population oracle weakly improves the ATE variance objective when the trial-only coefficient is feasible, but estimated coefficients do not inherit a universal no-harm property. The present 20-repetition studies suggest that the λ channel is the more stable source of gain and that joint tuning can add value in some designs. More Monte Carlo repetitions, completed CATE experiments, additional real- X supports, and an authorized real-outcome analysis are required before making a stronger empirical claim.

5.3. Scope, failure modes, and conclusion

The framework uses observational information through prediction rather than by declaring the observational treatment contrast causal. Trial randomization validates the pseudo-outcome for every fixed outcome regression. A common covariate marginal or an exact density ratio centers the ATE control variate and the prediction-powered CATE loss, which creates separate λ - and ω -channels while retaining RCT-population targets.

The first boundary is covariate transport. Without weighting under $P_R^X \neq P_O^X$, the ATE correction has drift $\omega\{P_O\hat{g} - P_R\hat{g}\}$, and the unweighted CATE loss targets the wrong covariate average. Exact $r = dP_R^X/dP_O^X$ removes both discrepancies. An estimated ratio instead contributes the explicit ATE drift $\omega P_O\{(\hat{r} - r)\hat{g}\}$ and the CATE validation drift in item 3 of Theorem 6. Evaluation-only variance is first-order adequate only under the stated negligible-drift conditions; otherwise ratio-training influence and covariance terms are required.

The second boundary is trial propensity overlap. Proposition 1 uses the known, strictly positive randomized-trial propensity. Extreme inverse-propensity factors can make the required moments unstable even when the conditional-mean identity remains algebraically valid, and an incorrect propensity removes the exact arbitrary- q guarantee.

The third boundary is weak observational prediction. The unrestricted ω -gain depends on how strongly $\hat{g}(X)$ predicts $\tau(X)$, with attenuation from the finite n/N ratio. The λ -channel gives little gain when $\Delta = Z_1 - Z_0$ is uninformative or nearly degenerate. Population oracles diagnose these limits but do not make their empirical coefficient estimates stable.

The fourth boundary is adaptive reuse. Fitting nuisances, tuning candidates, and evaluating the selected candidate on the same outcomes invalidates the conditioning product law. Present inference therefore uses independent nuisance-fitting, coefficient-tuning, and final-evaluation samples, supplied separately or constructed by splitting. The theory does not cover cross-fitted reuse; that extension requires a separate stability or influence-function analysis.

The fifth boundary is learner restriction and curvature. Target preservation does not make a finite-sample restricted or highly penalized projection accurate. The square-completed representation is a nonnegative mixture only for $0 \leq \omega < 1$, and positive semidefiniteness at $\omega = 1$ must be read from the original quadratic. Coefficients outside $[0, 1]$, nonlinear parameterizations, and penalty rescaling require direct analysis of the original objective.

The sixth boundary is tail behavior. ATE variance identities require square-integrable scores; the Wald theorem additionally requires conditional Lindeberg control, a sufficient $(2 + \delta)$ -moment condition, and a nondegenerate variance limit. Fixed-loss variance tuning imposes fourth-moment-type requirements. Heavy tails can therefore destabilize oracle formulas even when their algebraic denominators are nonzero.

Within these boundaries, the randomized trial fixes the causal target, the OBS outcome model can reduce within- X pseudo-outcome noise through λ , and the OBS covariate sample can reduce predictable between- X variation through ω . The exact results are conditional identities and population-oracle comparisons. Empirical work must retain the identical trial-only fallback, report selection and denominator instability, and avoid interpreting them as a universal finite-sample no-harm guarantee.

References

- Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. (2023). Prediction-powered inference. *Science*, 382(6671):669–674.
- Angelopoulos, A. N., Duchi, J. C., and Zrnic, T. (2024). PPI++: Efficient prediction-powered inference. *Journal of Machine Learning Research*, 25(104):1–59.
- Bareinboim, E. and Pearl, J. (2016). Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences*, 113(27):7345–7352.
- Cheng, D. and Cai, T. (2021). Adaptive combination of randomized and observational data. *Journal of the American Statistical Association*, 116(536):1863–1874.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2021). De-biased machine learning of global and local parameters using regularized riesz representers.
- Chernozhukov, V., Newey, W. K., and Singh, R. (2022). Automatic debiased machine learning of causal and structural effects. *Econometrica*, 90(3):967–1027.
- Colnet, B., Mayer, I., Chen, G., Dieng, A., Li, R., Varoquaux, G., Vert, J.-P., Josse, J., and Yang, S. (2020). Causal inference methods for combining randomized trials and observational studies: a review. *arXiv preprint arXiv:2011.08047*.
- Foster, D. J. and Syrgkanis, V. (2023). Orthogonal statistical learning. *The Annals of Statistics*, 51(3):879–908.
- Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46.
- Imai, K. and Ratkovic, M. (2014). Covariate balancing propensity score. *Journal of the Royal Statistical Society: Series B*, 76(1):243–263.
- Kallus, N., Puli, A. M., and Shalit, U. (2018). Removing hidden confounding by experimental grounding. *Advances in Neural Information Processing Systems*, 31.
- Kennedy, E. H. (2023). Towards optimal doubly robust estimation of heterogeneous causal effects. *Electronic Journal of Statistics*, 17(2):3008–3049.
- Künzel, S. R., Sekhon, J. S., Bickel, P. J., and Yu, B. (2019). Metalearners for estimating heterogeneous treatment effects using machine learning. *Proceedings of the National Academy of Sciences*, 116(10):4156–4165.
- Li, X., Li, Y., Cui, Q., Li, L., and Zhou, J. (2022). Robust direct learning for causal data fusion. In *Proceedings of the 14th Asian Conference on Machine Learning*, volume 189 of *Proceedings of Machine Learning Research*.
- Nie, X. and Wager, S. (2021). Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika*, 108(2):299–319.

- Rosenman, E., Basse, G., Owen, A., and Baiocchi, M. (2023). Combining observational and experimental datasets using shrinkage estimators. *Biometrics*, 79(4):2961–2973.
- Shi, X., Pan, Z., and Miao, W. (2023). Data integration in causal inference. *WIREs Computational Statistics*, 15(1):e1581.
- van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge University Press.
- Yang, S., Gao, C., Zeng, D., and Wang, X. (2023). Elastic integrative analysis of randomised trial and real-world data for treatment heterogeneity estimation. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 85(3):575–596.
- Yang, S., Liu, S., Zeng, D., and Wang, X. (2025). Data fusion methods for the heterogeneity of treatment effect and confounding function. *Bernoulli*, 31(4):2987–3012.

A. Omitted Preliminaries

A.1. Prediction-powered inference and PPI⁺⁺

Let (X_i, Y_i) , $i = 1, \dots, n$, be independent and identically distributed labeled observations from a distribution P , and let $\tilde{X}_j$, $j = 1, \dots, N$, be independent and identically distributed unlabeled covariates from the marginal distribution P^X , independently of the labeled sample. Let f be a fixed predictor. For a fixed coefficient $\omega \in \mathbb{R}$, the PPI⁺⁺ estimator ([Angelopoulos et al., 2024](#)) of the labeled-population mean is

$$\hat{\mu}_{\text{PPI}^{++}}(\omega) \triangleq \frac{1}{n} \sum_{i=1}^n Y_i + \omega \left\{ \frac{1}{N} \sum_{j=1}^N f(\tilde{X}_j) - \frac{1}{n} \sum_{i=1}^n f(X_i) \right\}. \quad (79)$$

If Y and $f(X)$ are integrable, then $\hat{\mu}_{\text{PPI}^{++}}(\omega)$ is unbiased for $\mathbb{E}(Y)$ for every fixed ω , because the labeled and unlabeled predictions have the same expectation. The same conclusion holds conditionally when ω is selected using data independent of the two samples displayed above. PPI⁺⁺ selects ω to control how strongly this mean-zero prediction difference enters the estimator. When $\text{Cov}\{Y, f(X)\} \neq 0$ and ω is close to its variance-minimizing value, the prediction difference can reduce variance relative to the labeled-only estimator.

B. Omitted Results

B.1. Convexity of the fusion losses

Step 2 of Algorithm 2 minimizes the losses of Definition 15 over $\zeta \in \mathcal{Z}$. The following expansion shows that this is a convex least-squares problem. It is recorded here because the results of Section 4.3 condition on the fitted function and do not use it.

Proposition 14 (Convexity of the fusion losses). Fix $j \in \{\text{DRF}, \text{RF}\}$, $\lambda \in [0, 1]$, $\omega \in [0, 1]$, and $r \geq 0$. Expanding the squares in Definition 15, $\hat{L}_j(\zeta; \lambda, \omega, r)$ is a quadratic functional of ζ whose second-order part is

$$\begin{aligned} \text{DRF: } & (1 - \omega) \mathbb{P}_R^{\text{tune}} \{\zeta(X)^2\} + \omega \mathbb{P}_O^{\text{tune}} \{r(X)\zeta(X)^2\}, \\ \text{RF: } & (1 - \omega) \mathbb{P}_R^{\text{tune}} \{\chi(A, X)\zeta(X)^2\} + \omega \mathbb{P}_O^{\text{tune}} \{r(X)\zeta(X)^2\}. \end{aligned} \quad (80)$$

Both are nonnegative, so $\hat{L}_j(\cdot; \lambda, \omega, r)$ is convex on any convex $\mathcal{Z}$. The expected loss $\mathbb{E}\{\hat{L}_j(\zeta; \lambda, \omega, r)\}$ has the same second-order part with $\mathbb{E}_R$ and $\mathbb{E}_O$ in place of $\mathbb{P}_R^{\text{tune}}$ and $\mathbb{P}_O^{\text{tune}}$, and is convex as well.

Convexity is the condition under which, by Definition 6, the minimizers of the expected losses are exactly the zeros of the learning equations in Definition 16.

B.2. Closed-form noise-floor coefficients

A closed-form choice of λ minimizes the noise floors of Proposition 8 instead of the CATE risk of the fitted learner in Definition 17. It is recorded here because the validation route of Section 4.3 targets the fitted risk directly and also chooses ω .

Proposition 15 (Noise-floor coefficients). Suppose Assumptions 2 and 1 hold and the displayed moments are finite. Recall Z_0 , Z_1 , and $\Delta = Z_1 - Z_0$ from Section 2.1, so that $Z_\lambda = Z_0 + \lambda\Delta$, and write $d_m \triangleq m_O - m_R$, so that $m_\lambda = m_R + \lambda d_m$. Let $\text{Proj}_{[0,1]}(u) \triangleq \min\{1, \max(0, u)\}$. If $\mathbb{E}_R(\Delta^2) > 0$, then

$$\arg \min_{\lambda \in [0,1]} Q_{\text{DR}}(\lambda) = \text{Proj}_{[0,1]} \left\{ -\frac{\mathbb{E}_R(Z_0\Delta)}{\mathbb{E}_R(\Delta^2)} \right\}, \quad (81)$$

and if $\mathbb{E}_R[d_m(X)^2 / \{e(X)(1 - e(X))\}] > 0$, then

$$\arg \min_{\lambda \in [0,1]} Q_R(\lambda) = \text{Proj}_{[0,1]} \left[\frac{\mathbb{E}_R[\{Y - m_R(X)\}d_m(X) / \{e(X)(1 - e(X))\}]}{\mathbb{E}_R[d_m(X)^2 / \{e(X)(1 - e(X))\}]} \right]. \quad (82)$$

If a denominator is zero, the corresponding noise floor is constant in λ . Both minimizers can be estimated by replacing the moments with tuning-sample averages.

C. Simulation Details

All completed simulation artifacts were generated with base seed 190602. The primary nested benchmarks use 20 outer Monte Carlo repetitions per design cell and compare four estimators: RCT-only AIPW, λ -only outcome-model fusion, ω -only PPI correction, and joint (λ, ω) fusion. The implementation retains the historical internal codes `haipw_only` and `ppi_power_tuned_only` for the second and third estimators, respectively. These codes are not the user-facing method names, and the first does not refer to the separate HAIPW paper.

The arbitrary simulation family uses three nonlinear structural causal models. In every model, a domain indicator S distinguishes the observational sample ($S = 0$) from the randomized sample ($S = 1$); covariates depend on latent variables, observational treatment depends on both measured covariates and an unmeasured confounder, trial treatment follows the known randomized propensity, and outcomes may depend nonlinearly on covariates, treatment, and the latent confounder. The common-marginal regime uses $P_R^X = P_O^X$. The shifted regime changes the covariate distribution while preserving overlap and estimates the density ratio from a domain classifier. The same estimators and evaluation logic are used in both regimes.

The fixed-OBS-budget experiment has 16 cells: four benchmark families, two covariate regimes, and $n_R \in \{20, 30\}$, with $N_O = 5000$. The proportional experiment has 32 cells: four benchmark families, two regimes, and $n_R \in \{20, 30, 50, 100\}$, with the exact mapping $N_O = 100n_R \in \{2000, 3000, 5000, 10000\}$. Coefficients are estimated by the nested honest procedure; no oracle coefficient or oracle density ratio is used in the reported method comparison. The replication-level CSV files, summaries, and composite plots in `materials/` are the canonical numerical artifacts.

The fourth benchmark family reuses the empirical covariate support from the Tennessee Student/Teacher Achievement Ratio (STAR) experiment. Treatment, latent confounding, outcomes, and treatment-effect heterogeneity are generated by the benchmark code. Consequently this is a real- X , semi-synthetic design, not an evaluation of the original STAR outcomes or intervention effect.

Figure 3 records the earlier two-method pilot. It is retained for provenance but is not the basis for the four-method conclusions in Section 5.

No participant-level WHI CT+OS analysis is included. Such an analysis requires authorized data access, a verified trial-observational semantic crosswalk, a prespecified estimand, and methods appropriate for censoring if the selected outcome is time-to-event. No completed CATE experiment or real-outcome result is represented by the present artifacts.

Illustrative random-SCM debugging pilot (20 outer refits per cell)
per stage: RCT n / OBS 5000; total generated per replicate: $3n / 15000$

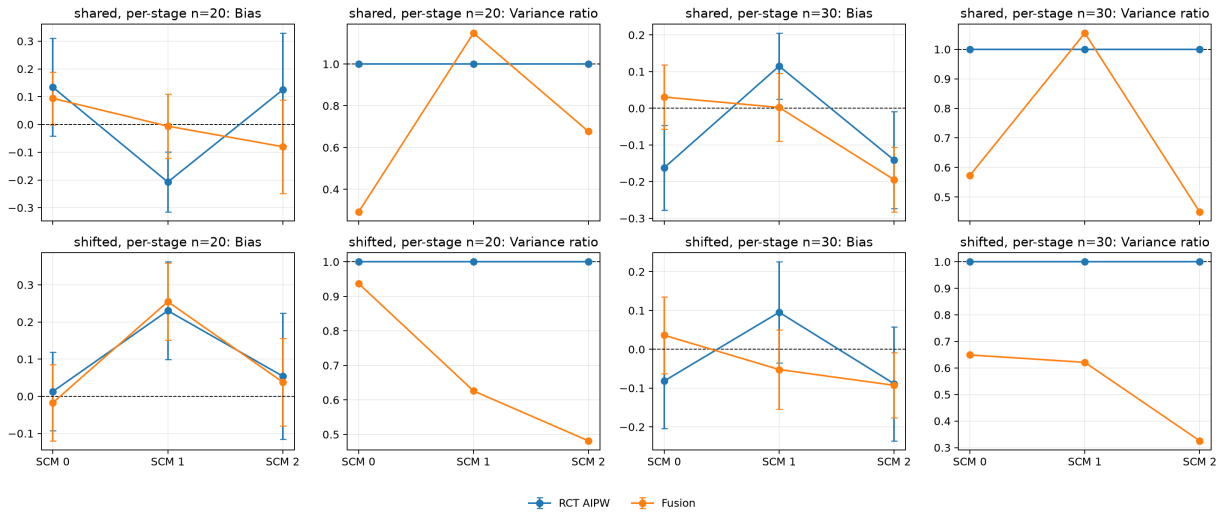


Figure 3: Earlier SSEM ATE pilot. This artifact predates the nested four-method comparison and should be interpreted only as a design and implementation check.

D. Proof Details

Unless stated otherwise, all proofs use Assumption 1: every nuisance, ratio, candidate function, and coefficient is fixed before the independent RCT and OBS evaluation observations are used.

D.1. Conditional validity of the trial pseudo-outcome

Fix $q = (q_0, q_1)$ without using the evaluation samples and condition on X . Under trial randomization, consistency, and honest sample separation,

$$\mathbb{E}_R \left[\frac{A}{e(X)} \{Y - q_1(X)\} \mid X \right] = \mathbb{E}_R \{Y(1) - q_1(X) \mid X\}, \quad (83)$$

$$\mathbb{E}_R \left[\frac{1 - A}{1 - e(X)} \{Y - q_0(X)\} \mid X \right] = \mathbb{E}_R \{Y(0) - q_0(X) \mid X\}. \quad (84)$$

Substituting these two equalities into the definition of φ cancels the two fitted regressions and yields $\mathbb{E}_R \{\varphi(V; q) \mid X\} = \tau(X)$. Applying the identity to $q = \hat{\mu}_R$, $q = \hat{\mu}_O$, and their affine blend gives

$$\mathbb{E}_R(Z_\lambda \mid X) = \tau(X), \quad \mathbb{E}_R(\Delta \mid X) = 0. \quad (85)$$

D.2. Proof of Corollary 1

Apply Equation (6) with the nuisance-sample fitted pair $q = \hat{\mu}_R$, average over D_R^{eval} , and use iterated expectation. This gives

$$\mathbb{E}\{\hat{\theta}_{\text{AIPW}, R}\} = \mathbb{E}_R\{\tau(X)\} = \theta.$$

D.3. Proof of Theorem 1

Pseudo-outcome validity and exact transport give

$$P_R(Z_\lambda) = \theta, \quad P_O(r_0 \hat{g}) = P_R(\hat{g}).$$

Substituting these identities into Equation (15) proves Equation (16). For any fixed $\tilde{r}$,

$$\begin{aligned} \mathbb{E}\{\hat{\theta}_{\tilde{r}}(\lambda, \omega)\} - \theta &= \omega \{P_O(\tilde{r} \hat{g}) - P_R(\hat{g})\} \\ &= \omega P_O\{(\tilde{r} - r_0) \hat{g}\}, \end{aligned}$$

which proves Equation (17). Using the same pseudo-outcome and transport identities, rewrite the centered estimator as

$$\hat{\theta}_{r_0}(\lambda, \omega) - \theta = (\mathbb{P}_{R,n} - P_R)\{Z_\lambda - \omega \hat{g}\} + (\mathbb{P}_{O,N} - P_O)\{\omega r_0 \hat{g}\}. \quad (86)$$

The two summands are independent because D_R^{eval} and D_O^{eval} are independent. The variance of an empirical mean of m independent observations is the population variance divided by m , which proves Equation (18). Unbiased sample variances on the two independent evaluation blocks therefore give the displayed fixed-selected-coefficient variance estimator in Section 3. Coverage of its

Wald interval additionally uses the stated evaluation-sample Lindeberg and moment assumptions; variance unbiasedness alone is not a central limit theorem.

D.4. Proof of Corollary 1.1

Expand the first variance in Equation (18) using $Z_\lambda = Z_0 + \lambda\Delta$. The interaction coefficient vanishes because

$$\begin{aligned}\text{Cov}_R(\Delta, \hat{g}) &= \mathbb{E}_R[\{\hat{g} - \mathbb{E}_R(\hat{g})\}\mathbb{E}_R(\Delta \mid X)] = 0, \\ D &= \text{Cov}_R(Z_0, \hat{g}) = \text{Cov}_R\{\mathbb{E}_R(Z_0 \mid X), \hat{g}\} = \text{Cov}_R\{\tau(X), \hat{g}(X)\}.\end{aligned}\quad (87)$$

Collecting the coefficient-dependent terms and multiplying by n gives $2C\lambda - 2D\omega + A\lambda^2 + B\omega^2$, which proves Equation (20). Finally, for every (u, v) ,

$$\begin{pmatrix} u & v \end{pmatrix} \mathbf{H}_{\text{ATE}} \begin{pmatrix} u \\ v \end{pmatrix} = Au^2 + Bv^2 \geq 0,$$

so $\mathbf{H}_{\text{ATE}}$ is positive semidefinite.

D.5. Proof of Proposition 2

Equation (20) is the positive affine transformation

$$\text{Var}\{\hat{\theta}_{r_0}(\lambda, \omega)\} = \frac{1}{n} \text{Var}_R(Z_0) + \frac{1}{n} \mathcal{J}_{\text{ATE}}(\lambda, \omega),$$

where $n > 0$. Adding a constant and multiplying by a positive scalar preserve the complete argmin set over every feasible set.

D.6. Proof of Theorem 2

When $A, B > 0$, the two first-order conditions are

$$2C + 2A\lambda = 0, \quad -2D + 2B\omega = 0.$$

The separated quadratic is strictly convex, so their unique solution is $\lambda^* = -C/A$ and $\omega^* = D/B$. For every $\rho \geq 0$, the ridge objective has gradient

$$\begin{pmatrix} 2C + 2(A + \rho)\lambda \\ -2D + 2(B + \rho)\omega \end{pmatrix}.$$

Since $A, B > 0$, both $A + \rho$ and $B + \rho$ are positive for every $\rho \geq 0$. Setting the gradient to zero therefore proves Equation (21) and uniqueness. At $\rho = 0$, the penalized objective reduces to $\mathcal{J}_{\text{ATE}}$, whose minimizers are also the minimizers of the exact population variance by Proposition 2.

D.7. Compact constraints

Continuity of $\mathcal{J}_{\text{ATE}}$ guarantees a minimizer on every nonempty compact rectangle. If the rectangle contains the origin, its minimum variance cannot exceed the RCT-only variance.

D.8. Proof of Corollary 2.1

For $A, B > 0$, complete the two scalar squares:

$$\mathcal{J}_{\text{ATE}}(\lambda, \omega) = A \left(\lambda + \frac{C}{A} \right)^2 + B \left(\omega - \frac{D}{B} \right)^2 - \frac{C^2}{A} - \frac{D^2}{B}.$$

The squared terms vanish at the oracle coefficients in Equation (22). Substitution into Equation (20) proves Equation (24).

D.9. Prediction-powered CATE losses and finite-sample theory

Proof of Proposition 8. Fix ζ, λ, ω and the fitted objects before using the tuning samples, and let m be m_λ or any fixed function. By Equation (6), $\mathbb{E}_R\{Z_\lambda - \tau(X) \mid X\} = 0$, so the cross term in $\mathbb{E}_R[\{Z_\lambda - \tau(X)\} + \{\tau(X) - \zeta(X)\}]^2$ vanishes and $\mathcal{L}_{\text{DR}}(\zeta; \lambda) = Q_{\text{DR}}(\lambda) + \mathcal{R}(\zeta)$. For the R loss, write $Y - m(X) - \{A - e(X)\}\zeta(X) = U + \{A - e(X)\}\{\tau(X) - \zeta(X)\}$ with $U \triangleq Y - m(X) - \{A - e(X)\}\tau(X)$. Conditional randomization gives $\mathbb{E}_R[\{A - e(X)\}\{Y - m(X)\} \mid X] = e(X)\{1 - e(X)\}\tau(X)$ and $\mathbb{E}_R[\{A - e(X)\}^2 \mid X] = e(X)\{1 - e(X)\}$, hence $\mathbb{E}_R[U\{A - e(X)\} \mid X] = 0$. Dividing the expanded square by $e(X)\{1 - e(X)\}$ and taking $\mathbb{E}_R(\cdot \mid X)$ therefore leaves $\mathcal{L}_R(\zeta; m) = Q_R(m) + \mathcal{R}(\zeta)$. Finally, exact transport gives $\mathbb{E}_O\{r_0(X)G_\zeta(X)\} = \mathbb{E}_R\{G_\zeta(X)\}$, and $\mathbb{E}_R\{\chi(A, X) \mid X\} = 1$ gives $\mathbb{E}_R\{\chi(A, X)G_\zeta(X)\} = \mathbb{E}_R\{G_\zeta(X)\}$, so both OBS corrections have mean zero for every ω . This proves Equation (56).

D.10. Proofs for coefficient selection

For Proposition 14, expand the squares in Equations (54) and (55). In the DRF loss, $\{Z_\lambda - \zeta(X)\}^2$ contributes $\zeta(X)^2$, and $G_\zeta(X) = \{\hat{g}(X) - \zeta(X)\}^2$ contributes $\zeta(X)^2$ to each term of the OBS bracket, so the second-order part is $(1 - \omega)\mathbb{P}_R^{\text{tune}}\{\zeta(X)^2\} + \omega\mathbb{P}_O^{\text{tune}}\{r(X)\zeta(X)^2\}$. In the RF loss, $[Y - m_\lambda(X) - \{A - e(X)\}\zeta(X)]^2/[e(X)\{1 - e(X)\}]$ contributes $\chi(A, X)\zeta(X)^2$, which gives the RF part of Equation (80). Since $\chi \geq 0$, $r \geq 0$, and $\omega \in [0, 1]$, both parts are nonnegative, and a quadratic functional with a nonnegative second-order part is convex on any convex set. Taking expectations over the tuning samples with the fitted objects fixed replaces $\mathbb{P}_R^{\text{tune}}$ and $\mathbb{P}_O^{\text{tune}}$ by $\mathbb{E}_R$ and $\mathbb{E}_O$, which proves the claim for the expected losses.

For Proposition 15, pseudo-outcome validity gives $\mathbb{E}_R(\Delta \mid X) = 0$. Therefore

$$\begin{aligned} Q_{\text{DR}}(\lambda) &= Q_{\text{DR}}(0) + 2\lambda\mathbb{E}_R[(Z_0 - \tau)\Delta] + \lambda^2\mathbb{E}_R(\Delta^2), \\ \mathbb{E}_R[(Z_0 - \tau)\Delta] &= \mathbb{E}_R(Z_0\Delta). \end{aligned}$$

Completing the square gives Equation (81) when $\mathbb{E}_R(\Delta^2) > 0$. If $\mathbb{E}_R(\Delta^2) = 0$, then $\Delta = 0$ almost surely and the objective is flat.

For the R loss, write $U_R \triangleq Y - m_R - (A - e)\tau$. Since $m_\lambda = m_R + \lambda d_m$,

$$\begin{aligned} Q_R(\lambda) &= Q_R(0) - 2\lambda \mathbb{E}_R(U_R d_m / \{e(1 - e)\}) + \lambda^2 \mathbb{E}_R(d_m^2 / \{e(1 - e)\}), \\ \mathbb{E}_R(U_R d_m / \{e(1 - e)\}) &= \mathbb{E}_R[\{Y - m_R\} d_m / \{e(1 - e)\}]. \end{aligned}$$

The second identity follows from $\mathbb{E}_R(A - e \mid X) = 0$. Completing the square proves Equation (82). If $\mathbb{E}_R(d_m^2 / \{e(1 - e)\}) = 0$, strict overlap implies $d_m = 0$ almost surely and the objective is flat. A convex quadratic minimized over $[0, 1]$ is minimized by projecting its unrestricted minimizer onto that interval.

For Theorem 6, condition on the fitted objects, which are independent of the evaluation samples by Assumption 1. Item 1: by Equation (6), $\mathbb{E}_R\{Z_0 - \hat{\zeta}_j(X; \lambda, \omega)\}^2 = Q_{\text{DR}}(0) + \mathcal{R}\{\hat{\zeta}_j(\cdot; \lambda, \omega)\}$ as in the proof of Proposition 8, and exact transport gives $\mathbb{E}_O(r_0 G_{j, \lambda, \omega}) = \mathbb{E}_R(G_{j, \lambda, \omega})$, so the correction has mean $\omega \mathbb{E}_O(r G_{j, \lambda, \omega}) - \omega \mathbb{E}_R(G_{j, \lambda, \omega}) = \omega \mathbb{E}_O[(r - r_0) G_{j, \lambda, \omega}]$. Item 2: each RCT summand $\{Z_0 - \hat{\zeta}_j\}^2 - \omega G_{j, \lambda, \omega}$ lies in $[-4B^2, 4B^2]$ and each OBS summand $\omega r G_{j, \lambda, \omega}$ in $[0, 4\bar{r}B^2]$, so Hoeffding's inequality bounds the deviation of the RCT average by $8B^2 \sqrt{\log(4M/\alpha)/(2n)}$ and that of the OBS average by $4\bar{r}B^2 \sqrt{\log(4M/\alpha)/(2N)}$, each with probability at least $1 - \alpha/(2M)$. A union bound over the M grid pairs and the two averages gives both bounds for every pair with probability at least $1 - \alpha$, and the two deviations sum to $\varepsilon_{\text{eval}}(\alpha)$. Item 3: write Q for $Q_{\text{DR}}(0)$, $b(\lambda, \omega)$ for the transport error in item 1, $\hat{s}$ for the selected pair, and s^* for a grid pair minimizing the CATE risk. On the event of item 2, $\mathcal{R}(\hat{s}) = \mathbb{E}\{\hat{C}_j(\hat{s})\} - Q - b(\hat{s}) \leq \hat{C}_j(\hat{s}) - Q - b(\hat{s}) + \varepsilon_{\text{eval}}(\alpha) \leq \hat{C}_j(s^*) - Q - b(\hat{s}) + \varepsilon_{\text{eval}}(\alpha) \leq \mathcal{R}(s^*) + b(s^*) - b(\hat{s}) + 2\varepsilon_{\text{eval}}(\alpha)$, where the middle inequality holds because $\hat{s}$ minimizes $\hat{C}_j$ over the grid. Bounding $|b(s^*) - b(\hat{s})|$ by twice the maximum of $|b|$ over the grid proves Equation (66).

D.11. Proofs for the orthogonality audit

Condition throughout on the nuisance samples, so that $\hat{\mu}_R$, $\hat{\mu}_O$, $\hat{g}$, r , λ , and ω are fixed before the tuning samples are used, and assume enough integrability to interchange differentiation and expectation.

Learning equations. Differentiating the expected losses of Definition 15 in the direction v and multiplying by $-\frac{1}{2}$ gives Equations (58) and (59). For $r = r_0$, exact transport gives $\mathbb{E}_O(r_0 f) = \mathbb{E}_R(f)$ for each integrable $f(X)$, and Assumption 2 gives $\mathbb{E}_R\{\chi(A, X) \mid X\} = 1$, so the two terms of each OBS bracket cancel, which gives the reduced forms displayed after Definition 16.

Proof of Proposition 9. For a pair $h = (h_0, h_1)$ write $c(h) \triangleq \{A - e(X)\}\{h_1(X)/e(X) + h_0(X)/[1 - e(X)]\}$. Differentiating $\varphi(V; q)$ in Definition 2 along $q + \varepsilon h$ gives $-c(h)$, and $\mathbb{E}_R\{c(h) \mid X\} = 0$ by conditional randomization. Along $\hat{\mu}_R + \varepsilon h$, the fused pair $\hat{\mu}_\lambda$ moves by $(1 - \lambda)h$ and m_λ moves by $(1 - \lambda)\{e(X)h_1(X) + [1 - e(X)]h_0(X)\}$, while $\hat{g}$ does not move. Hence

$$\begin{aligned} D_{\hat{\mu}_R} \Psi_{\text{DRF}}[h, v] &= -(1 - \lambda) \mathbb{E}_R\{v(X) c(h)\} = 0, \\ D_{\hat{\mu}_R} \Psi_{\text{RF}}[h, v] &= -(1 - \lambda) \mathbb{E}_R \left[v(X) \frac{A - e(X)}{e(X)\{1 - e(X)\}} \{e(X)h_1(X) + [1 - e(X)]h_0(X)\} \right] = 0, \end{aligned}$$

because v and h are functions of X and $\mathbb{E}_R\{A - e(X) \mid X\} = 0$. This proves item 1. Along $\hat{\mu}_O + \varepsilon h$, the same two contributions appear with λ in place of $1 - \lambda$ and vanish for the same

reason, and in addition $\hat{g}$ moves by $u = h_1 - h_0$. Only the OBS bracket depends on $\hat{g}$, and its derivative is $\omega[\mathbb{E}_O\{r(X)v(X)u(X)\} - \mathbb{E}_R\{v(X)u(X)\}]$ for DRF and $\omega[\mathbb{E}_O\{r(X)v(X)u(X)\} - \mathbb{E}_R\{\chi(A, X)v(X)u(X)\}]$ for RF. Exact transport and $\mathbb{E}_R\{\chi(A, X) \mid X\} = 1$ turn both into $\omega\mathbb{E}_O[\{r(X) - r_0(X)\}u(X)v(X)]$, which proves item 2. Along $r + \varepsilon h_r$, only the positive OBS term moves, so $D_r\Psi_j[h_r, v] = \omega\mathbb{E}_O[h_r(X)v(X)\{\hat{g}(X) - \zeta(X)\}]$. For the last claim of item 3, take $f \triangleq v(\hat{g} - \zeta) \in L_2(P_O^X)$. Its inner product with every mean-zero element of $L_2(P_O^X)$ is zero if and only if f lies in the orthogonal complement of the mean-zero functions, which consists of the almost-sure constants; for example, the centered direction $h_r = f - \mathbb{E}_O(f)$ has pairing $\mathbb{E}_O(h_r f) = \text{Var}_O(f)$. Multiplication by nonzero ω does not change the zero set.

For Proposition 10, expand each quadratic objective in β . The DRF curvature and right-hand side are $(1 - \omega)\hat{\Gamma}_R + \omega\hat{\Gamma}_{O,r} + \rho P$ and $\hat{h}_\lambda + \omega\hat{d}_{\text{OBS}}$. The RF counterparts are $(1 - \omega)\hat{\Gamma}_{R,\chi} + \omega\hat{\Gamma}_{O,r} + \rho P$ and $\hat{h}_{m_\lambda} + \omega\hat{d}_{\text{OBS},\chi}$. Setting the gradients to zero proves Equations (67) and (68). For $0 \leq \omega \leq 1$, each unpenalized Hessian is a convex combination of positive-semidefinite weighted Gram matrices; adding $\rho P \succeq 0$ preserves positive semidefiniteness.

For Proposition 11, fixed ω makes the Hessian independent of λ , while $Z_\lambda = Z_0 + \lambda\Delta$ and $m_\lambda = m_R + \lambda d_m$ make the right-hand side affine. Multiplication by the fixed inverse Hessian proves Equation (69). Substitution of the affine fitted function into the separate tuning criterion and expansion of its squares proves Equation (70). The coefficients are the constant, linear, and quadratic terms of that expansion.

For Theorem 7, the first-order condition is $(\hat{H}_j + \rho P)\hat{\beta}_j = \hat{h}_j$. Subtracting its value at β_p proves Equation (71). Projection orthogonality, $\mathbb{E}_R[b(\zeta_p - \tau)] = 0$, removes the risk cross term and proves Equation (72).

For Theorem 8, let $\psi_{R,j}$ and $\psi_{O,j}$ be the explicit raw contributions in the main text. Their source means need not vanish separately, but exact transport gives $P_R\psi_{R,j} + P_O\psi_{O,j} = 0$. Center each source by its own mean. Since the centered RCT and OBS sample means are independent, their covariance proxies add:

$$\text{Cov}(\mathbb{P}_R\psi_{R,j} + \mathbb{P}_O\psi_{O,j}) = \frac{\Sigma_{R,j}}{n_f} + \frac{\Sigma_{O,j}}{N_f}.$$

The stated sub-Gaussian vector concentration inequality bounds the squared $\Gamma_R^{-1/2}$ -norm by the sum of the two trace terms and their operator norm tail terms, without applying a triangle inequality between sources. On $\mathcal{E}_H$, the whitened inverse penalized Hessian has norm at most $(1 - \eta)^{-1}$. Combining this fact with Equation (71) and bounding the ridge vector separately proves Equation (73).

For Proposition 12, write the leading term as $A_{n_f, N_f} \triangleq \Gamma_R^{-1}\{(\mathbb{P}_R - P_R)\psi_{R,j} + (\mathbb{P}_O - P_O)\psi_{O,j}\}$, so the remainder defined in the proposition gives $\hat{\beta}_j - \beta_p = A_{n_f, N_f} + R_{n_f, N_f}$. Insert this identity into Equation (72). Independence removes the cross covariance between the RCT and OBS components of A_{n_f, N_f} , and hence

$$\mathbb{E}\|A_{n_f, N_f}\|_{\Gamma_R}^2 = \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{R,j})}{n_f} + \frac{\text{tr}(\Gamma_R^{-1}\Sigma_{O,j})}{N_f}.$$

Moreover, Cauchy–Schwarz and the stated mean-square remainder condition give

$$\left| \mathbb{E} \langle A_{n_f, N_f}, R_{n_f, N_f} \rangle_{\Gamma_R} \right| \leq \{ \mathbb{E} \|A_{n_f, N_f}\|_{\Gamma_R}^2 \}^{1/2} \{ \mathbb{E} \|R_{n_f, N_f}\|_{\Gamma_R}^2 \}^{1/2} = o(n_f^{-1} + N_f^{-1}),$$

and the expected squared remainder has the same smaller order. The identities $\text{tr}(AB) = \text{tr}(BA)$ then give Equation (74). The comparison in Corollary 8.1 follows by subtracting the two leading expansions; it is not a finite-sample comparison.

For Proposition 13, write the estimated- ratio normal equation as $(\hat{H}_j^{r_0} + \hat{\Delta}_{H,j} + \rho P) \hat{\beta}_j^r = \hat{h}_j^{r_0} + \hat{\Delta}_{h,j}$. Subtract the exact-ratio equation and rearrange. Invertibility gives the displayed resolvent identity.

Finally, substitute $a_K^2 = O(K^{-2s})$ and the covariance scales

$$\frac{K c_{R,j}(K; \lambda, \omega)}{n_f} \quad \text{and} \quad \frac{K c_{O,j}(K; \lambda, \omega)}{N_f}$$

into the finite-sample result. Adding the stated ridge and fitting-ratio terms proves Equation (78). Minimizing the resulting criterion over $\mathcal{K}_n$ gives the displayed definition of K_j^* . Under the stated uniform positive finite limits, that criterion is asymptotically equivalent, uniformly over the coefficient candidates, to $K^{-2s} + K v_j^\infty$. Balancing its two terms gives $K \asymp (v_j^\infty)^{-1/(2s+1)}$ and rate $(v_j^\infty)^{2s/(2s+1)}$, proving Corollary 8.2.

D.12. A two-stratum oracle calculation

Let $P_R(X = 0) = P_R(X = 1) = 1/2$, $\tau(0) = 0$, $\tau(1) = 2$, and $\hat{g} = \tau$. With $n = 100$, $N = 900$, and a common covariate marginal,

$$\mathbb{E}_R\{\tau(X)\} = 1, \quad \text{Var}_R\{\tau(X)\} = 1, \quad \omega_{\text{ATE}}^* = \frac{1}{1 + 100/900} = 0.9.$$

The baseline between- X variance is 0.01, whereas the oracle component is

$$\frac{\text{Var}_R\{0.1\tau(X)\}}{100} + \frac{0.9^2 \text{Var}_R\{\tau(X)\}}{900} = 0.0001 + 0.0009 = 0.001.$$

The reduction 0.009 concerns only between- X variation. The first RCT variance term in Equation (18) also contains conditional pseudo-outcome noise.

D.13. Honest variance estimation and evaluation-sample inference

For Proposition 3, fix the coefficients and scores before using the evaluation observations. Ordinary unbiased sample-variance identities give

$$\mathbb{E}\{\widehat{\text{Var}}_{R,\text{eval}}(\psi_R)\} = \text{Var}_R(\psi_R), \quad \mathbb{E}\{\widehat{\text{Var}}_{O,\text{eval}}(\psi_O)\} = \text{Var}_O(\psi_O).$$

Dividing by n and N , then adding, proves unbiasedness of $\hat{V}$ by independence of D_R^{eval} and D_O^{eval} .

For Theorem 4, impose the following evaluation regularity conditions after fixing the learned objects:

- (R1) the RCT and OBS evaluation observations are conditionally independent and retain their respective product laws;
- (R2) $n, N \rightarrow \infty$, $n/N \rightarrow \kappa \in [0, \infty)$, and for some $\delta > 0$ the centered RCT and OBS evaluation-score arrays have uniformly bounded conditional $(2 + \delta)$ moments;
- (R3) $n \text{Var}\{\hat{\theta}_r(\hat{\lambda}, \hat{\omega})\} \rightarrow \sigma^2 \in (0, \infty)$ in probability, and the combined sample-variance estimator is ratio-consistent for this variance.

Center the two evaluation-score arrays at their population means. Their weighted sum is $\hat{\theta}_r - \theta$. Condition (R2) supplies the two-sample triangular-array central limit theorem: the RCT Lyapunov remainder is of order $n^{-\delta/2}$, and the OBS remainder also vanishes when $n/N \rightarrow \kappa < \infty$. Condition (R3) and Slutsky's theorem yield the studentized limit and asymptotic coverage of the normal-approximation interval.

D.14. Proof of Theorem 3

Under either exact $r = r_0$ transport or its common-marginal special case, Equation (6) gives $\mathbb{E}_R(\Delta \mid X) = 0$. Hence $\text{Cov}_R(\Delta, \hat{g}) = 0$, as shown in Equation (87). Multiplying Equation (20) by n therefore gives

$$\tilde{\mathcal{V}}(\lambda, \omega) = \sigma_0^2 + 2C\lambda - 2D\omega + A\lambda^2 + B\omega^2.$$

At $(\lambda, \omega) = (0, 0)$, this equals σ_0^2 . Holding $\lambda = 0$ and completing the square in ω gives the unique PPI⁺⁺-only oracle D/B and variance $\sigma_0^2 - D^2/B$. Completing the two separate squares gives the joint oracle $(-C/A, D/B)$ and variance $\sigma_0^2 - C^2/A - D^2/B$. The two variance gaps are D^2/B and C^2/A . Because $A, B > 0$, they are nonnegative and vanish exactly when $D = 0$ and $C = 0$, respectively.

D.15. Density-ratio characterizations and estimated-ratio inheritance

Proof of Proposition 4. The defining property of the Radon–Nikodym derivative gives

$$\mathbb{E}_O\{r_0(X)h(X)\} = \mathbb{E}_R\{h(X)\}$$

for every integrable h . If a nonnegative normalized function $\tilde{r}$ gives the same equality for every bounded measurable h , then $\int_A \tilde{r} dP_O = P_R(A) = \int_A r_0 dP_O$ for every measurable set A . Uniqueness of the Radon–Nikodym derivative gives $\tilde{r} = r_0$ P_O^X -almost surely.

Proof of Proposition 5. If $r_0 \in L_2(P_O^X)$, then for every $a \in L_2(P_O^X)$,

$$\begin{aligned} \mathcal{R}(a) &= \mathbb{E}_O\{a(X)^2\} - 2\mathbb{E}_R\{a(X)\} \\ &= \mathbb{E}_O\{(a(X) - r_0(X))^2\} - \mathbb{E}_O\{r_0(X)^2\}. \end{aligned}$$

The squared term is uniquely minimized at $a = r_0$ in $L_2(P_O^X)$. Restricting the balance equations to the coordinates of ϕ tests equality only against functions in their linear span, which proves the finite-feature qualification.

Proof of Lemma 1. For $a_\gamma = \phi^\top \gamma$, the gradient of the objective in Equation (34) is twice

$$\left[\mathbb{P}_O^{\text{nuis}} \{ \phi(X) \phi(X)^\top \} + \rho I_p \right] \gamma - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}.$$

The stated invertibility condition makes its unique zero Equation (35).

Classifier odds. Let p_R and p_O be densities with respect to a common dominating measure. In the pooled nuisance population,

$$s_0(x) = \frac{\pi p_R(x)}{\pi p_R(x) + (1 - \pi) p_O(x)}.$$

Solving for $p_R(x)/p_O(x)$ gives

$$r_0(x) = \frac{1 - \pi}{\pi} \frac{s_0(x)}{1 - s_0(x)},$$

which proves Equation (36) in Definition 11. The same algebra with a fitted domain probability and the effective training prior yields $\hat{r}_{\text{CLS}}$.

Proof of Proposition 6. The log of a positive weighted exponential sum is convex in ξ . At an interior minimizer, differentiating the objective in Definition 12 gives

$$\begin{aligned} 0 &= \frac{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \phi(X) \}}{\mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{base}}(X) e^{\hat{\xi}^\top \phi(X)} \}} - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \} \\ &= \mathbb{P}_O^{\text{nuis}} \{ \hat{r}_{\text{BAL}}(X) \phi(X) \} - \mathbb{P}_R^{\text{nuis}} \{ \phi(X) \}. \end{aligned}$$

The denominator normalizes the OBS weights to one, and positivity follows from the positive base ratio and exponential tilt. A finite interior optimizer can exist only when the target feature mean lies in the relative interior of the convex hull of the weighted OBS feature vectors. Boundary or infeasible cases require a separate approximate-balance rule and are not covered by the exact first-order condition.

Proof of Proposition 7. Item 1. On the tuning OBS sample, $\hat{r}\hat{g} = r_0\hat{g} + (\hat{r} - r_0)\hat{g}$, and the sample-variance identity $\widehat{\text{Var}}(U + V) = \widehat{\text{Var}}(U) + 2\widehat{\text{Cov}}(U, V) + \widehat{\text{Var}}(V)$ with $U = r_0\hat{g}$ and $V = (\hat{r} - r_0)\hat{g}$ proves Equation (41). The moments $\hat{A}$, $\hat{C}$, and $\hat{D}$ in Equation (28) do not involve the ratio, so the ratio enters Equation (30) only through $\hat{B}$.

Item 2. At the same selected coefficients, the two estimators in Equation (15) share the term $\mathbb{P}_{R,n} Z_{\hat{\gamma}} - \hat{\omega} \mathbb{P}_{R,n} \hat{g}$ and differ only in $\hat{\omega} \mathbb{P}_{O,N}(r\hat{g})$, whose difference across $r = \hat{r}$ and $r = r_0$ is $\hat{\omega} \mathbb{P}_{O,N} \{ (\hat{r} - r_0) \hat{g} \}$.

Item 3. Condition on all fitted and selected objects. With the evaluation scores of Algorithm 1, $\hat{\theta}_{\hat{r}} = \mathbb{P}_{R,n} \psi_R + \mathbb{P}_{O,N} \psi_O$. The exact drift identity in Equation (17) gives the conditional mean $\theta + \hat{\omega} \mathbb{E}_O \{ (\hat{r} - r_0) \hat{g} \}$, and the transport identity in Equation (32) rewrites $\mathbb{E}_O(r_0 \hat{g})$ as $\mathbb{E}_R(\hat{g})$. Independence of the two evaluation samples gives $\text{Var}(\hat{\theta}_{\hat{r}}) = \text{Var}_R(\psi_R)/n + \text{Var}_O(\psi_O)/N$, and each sample variance in $\hat{V}$ is conditionally unbiased for the corresponding variance, which proves Equation (43).

Proof of Theorem 5. Write $\beta \triangleq \hat{\omega}\{\mathbb{E}_O(\hat{r}\hat{g}) - \mathbb{E}_R(\hat{g})\}$ for the drift in item 3 of Proposition 7. Continue conditioning on the fitted and selected objects. Center the two evaluation scores at their respective conditional means. The resulting two independent triangular arrays sum to $\hat{\theta}_{\hat{r}} - (\theta + \beta)$ and have total variance V_n . On the event $nV_n \geq \sigma^2/2$, whose probability tends to one, the uniform conditional $(2+\delta)$ -moment bound and bounded n/N make the conditional Lyapunov ratio bounded by an $O_p(1)$ factor times

$$n^{-\delta/2} + \frac{n^{1+\delta/2}}{N^{1+\delta}} = o_p(1).$$

Thus its conditional law given the fitted and selected objects converges in probability to $N(0,1)$ after division by $\sqrt{V_n}$. This also implies the unconditional convergence.

It remains to establish rather than assume variance consistency. Let $\eta \triangleq \min(\delta/2, 1)$. For a centered score ξ , the conditional $(2+\delta)$ -moment bound gives $\mathbb{E}(|\xi^2 - \mathbb{E}\xi^2|^{1+\eta} \mid \text{fitted objects}) = O_p(1)$. The von Bahr–Esseen inequality therefore bounds the conditional $(1+\eta)$ -moment of the centered empirical second moment by $O_p(m^{-\eta})$ for a sample of size m . Moreover, the squared empirical mean is $o_p(1)$ because its conditional expectation is $O_p(m^{-1})$. Consequently, for $S \in \{R, O\}$,

$$\widehat{\text{Var}}_{S,\text{eval}}(\psi_S) - \text{Var}_S(\psi_S) = o_p(1).$$

Consequently, because n/N is bounded,

$$n\{\hat{V} - V_n\} = o_p(1) + \frac{n}{N}o_p(1) = o_p(1).$$

Together with $nV_n \rightarrow_p \sigma^2 > 0$, this proves $\hat{V}/V_n \rightarrow_p 1$. Finally,

$$\frac{\hat{\theta}_{\hat{r}} - \theta}{\sqrt{\hat{V}}} = \frac{\hat{\theta}_{\hat{r}} - (\theta + \beta)}{\sqrt{V_n}} \sqrt{\frac{V_n}{\hat{V}}} + \frac{\beta}{\sqrt{V_n}} \sqrt{\frac{V_n}{\hat{V}}}.$$

The bias condition and Slutsky's theorem prove the stated limit and coverage.

Proof of Corollary 5.1. Fix $k \in \{\text{RCT}, \text{PPI}^{++}, \text{joint}\}$ and write $(\hat{\lambda}, \hat{\omega}) \triangleq (\hat{\lambda}_k, \hat{\omega}_k)$ and $(\lambda^*, \omega^*) \triangleq (\lambda_k^*, \omega_k^*)$. Decompose

$$\hat{\theta}_{\hat{r},k} - \hat{\theta}_{r_0}(\lambda^*, \omega^*) = \left\{ \hat{\theta}_{\hat{r}}(\hat{\lambda}, \hat{\omega}) - \hat{\theta}_{r_0}(\hat{\lambda}, \hat{\omega}) \right\} + \left\{ \hat{\theta}_{r_0}(\hat{\lambda}, \hat{\omega}) - \hat{\theta}_{r_0}(\lambda^*, \omega^*) \right\}.$$

By item 2 of Proposition 7, the first difference equals $\hat{\omega}\mathbb{P}_{O,N}\{(\hat{r} - r_0)\hat{g}\}$. Its conditional mean is the drift of item 3, which the bias condition in Theorem 5 makes $o_p(n^{-1/2})$ because $nV_n = O_p(1)$. Its conditional variance is $\hat{\omega}^2 \text{Var}_O\{(\hat{r} - r_0)\hat{g}\}/N$, so Equation (48) and conditional Chebyshev make its centered part $o_p(n^{-1/2})$. For the second difference,

$$\hat{\theta}_{r_0}(\hat{\lambda}, \hat{\omega}) - \hat{\theta}_{r_0}(\lambda^*, \omega^*) = (\hat{\lambda} - \lambda^*)\mathbb{P}_{R,n}\Delta + (\hat{\omega} - \omega^*)\{\mathbb{P}_{O,N}(r_0\hat{g}) - \mathbb{P}_{R,n}\hat{g}\}.$$

RCT validity gives $P_R\Delta = 0$, and exact transport gives $P_O(r_0\hat{g}) = P_R(\hat{g})$. Together with the second-moment bounds in Equation (47) and bounded n/N , these identities make both empirical factors $O_p(n^{-1/2})$, and coefficient consistency makes the second difference $o_p(n^{-1/2})$. Combining the two differences proves Equation (49). Convergence of the exact-ratio second moments then transfers the ordering in Theorem 3, which is Equation (50).

D.16. Counterexamples to adaptive reuse

The fixed-coefficient theorem cannot be extended by selecting a coefficient on the same evaluation outcomes without a separate analysis.

Reuse through λ . Let X be degenerate, $\theta = 0$, $Z_0 = 0$, and let the Δ_i be i.i.d. Rademacher variables. Every fixed λ satisfies $\mathbb{E}\{\lambda \mathbb{P}_{R,n} \Delta\} = 0$. If the same outcomes select $\hat{\lambda} = \text{sign}(\mathbb{P}_{R,n} \Delta)$, then

$$\mathbb{P}_{R,n}(Z_0 + \hat{\lambda} \Delta) = |\mathbb{P}_{R,n} \Delta|,$$

whose expectation is strictly positive.

Reuse through ω . Under the common covariate marginal, let $\hat{g}(X)$ be a nondegenerate mean-zero Rademacher variable and let the RCT-valid score be identically zero. Every fixed ω centers $\omega\{\mathbb{P}_{O,N} \hat{g} - \mathbb{P}_{R,n} \hat{g}\}$. Choosing $\hat{\omega}$ as the sign of that same bracket produces its absolute value, which again has strictly positive expectation. The bias in both examples comes solely from adaptive reuse.